$\bar{F}_j \leftarrow -(\bar{B}_j - A_j \bar{C}_{j-1})^{-1} (F_j - A_j F_{j-1})$

$B_j \leftarrow I$

Last Row, N

$B_N \leftarrow I$

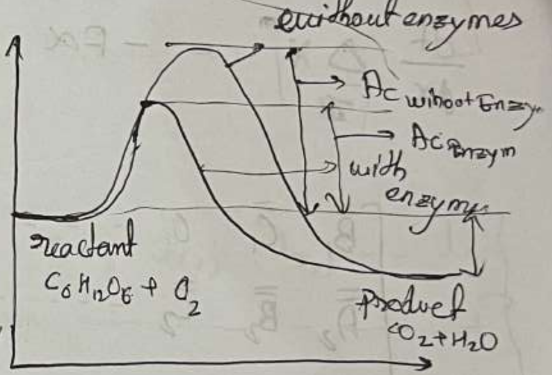
$A_N \leftarrow 0$, $C_N \leftarrow$ vanish

$F_N \leftarrow -(B_N - A_N C_{N-1})^{-1} (F_N - A_N F_{N-1})$

What are enzymes & its Role

↳ Enzymes are globular proteins which has one or many active sites, which enables selectives reaction to happens.

Enzymes lower the energy requirement of a reaction thereby speed up reaction rate.



Enzyme are present in — • Signal transmission in neurons
• Ion transport in cell membrane | • Harmonal synthes
• Energy production | • All physiological function.

(*) It take part in the reaction, in the intermediate steps, form some complex but catalyst only decrease the Activation energy. ↳ Reaction Intermediate.

(*) At least one Intermediate step must be present.

(*) There is always a reversible step present.

$$E + S \rightleftharpoons ES \longrightarrow P + E$$

↑ Enzyme ↑ Substrate

↳ always a reversible step present

Substrate $\longrightarrow$ Product
Glucose $\uparrow$ ATP, ADP
Enzyme

* In short, enzymes are catalysts that facilitate all biochemical reaction in living body.

Adsorption $\rightarrow$ No (product / transformation / conversion) happens.

$$\Delta G = \underbrace{\Delta H}_{\text{not changing}} - T\Delta S$$ $\rightarrow$ changing change

Entropy decreases.
(Because from disordered to ordered state)

* In Enzymatic Reaction.

$$S \xrightarrow[(X)]{} P$$

$$S + E \rightleftharpoons ES \longrightarrow P + E$$

$$A + B \xrightarrow{k} C$$

$$r = -\frac{d[A]}{dt} = -\frac{d[B]}{dt} = \frac{d[C]}{dt} = k(T)[A]^n[B]^m$$

for typical Enzymatic Reaction $\rightarrow$

Write reaction rates in terms of reactants only

$$E + S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} ES \xrightarrow{k_2} E + P$$

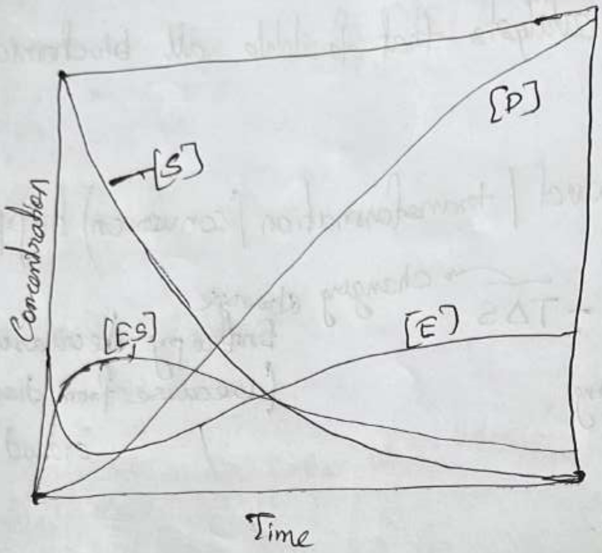
(1) $\frac{d[ES]}{dt} = k_1 [E][S] - k_{-1}[ES] - k_2[ES]$ IC 1 $[ES]=0$ @ $t=0$

Enzyme moles should be consumed $[E]_t + [ES]_t = (\text{const})$ $[E]_0$ — (2)

$\frac{d[S]}{dt} = -k_1[S][E] + k_{-1}[ES]$ — (3) IC 2 $[S] = S_0$ at $t=0$

$\frac{d[P]}{dt} = k_2[ES]$ — (4) IC 3 $[P]=0$ at $t=0$

four species $[E], [ES], [P], [S]$



[E] & [ES] are missing Image

for maximising ~~By~~ prd $\rightarrow \frac{d[ES]}{dt} = 0$

$$K_1[E][S] - K_{-1}[ES] - K_2[ES]$$

$$[ES] = \frac{K_1[E][S]}{K_{-1} + K_2}$$

$$\therefore \frac{d[S]}{dt} = -K_1[S][E] + K_{-1}[ES]$$ ~~at m~~

at max $[ES] =$

$$\frac{d[S]}{dt} = K_1[S][E]\left[\frac{K_{-1}}{K_{-1}+K_2} - 1\right] = \frac{-K_1K_2[S][E]}{K_{-1}+K_2}$$

$$\frac{d[P]}{dt} = \frac{K_2K_1[S][E]}{K_{-1}+K_2}$$

Influencing factor & self control of reaction rate

→ Temperature
→ Presence of inhibitor molecules

(*) Temperature → require optimum temp to work.

(*) Excess substrate, product, enzymes → in these cases reactions become self controlled, not that rate will increase/decrease

Substrate

$$E + S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} ES \xrightarrow{k_2} P$$

$$K_s \Updownarrow S$$

$$ESS$$

→ If excess substrate, then it will bind to ES complex to form ESS, so excessive substrate did not lead to excessive product (Natural control on the process)

ESS, EP, ESP → won't form product only enzyme-substrate complex (ES) can form product

Product

$$E + S \rightleftharpoons ES \rightarrow E + P$$

$$+P \qquad\qquad +P$$

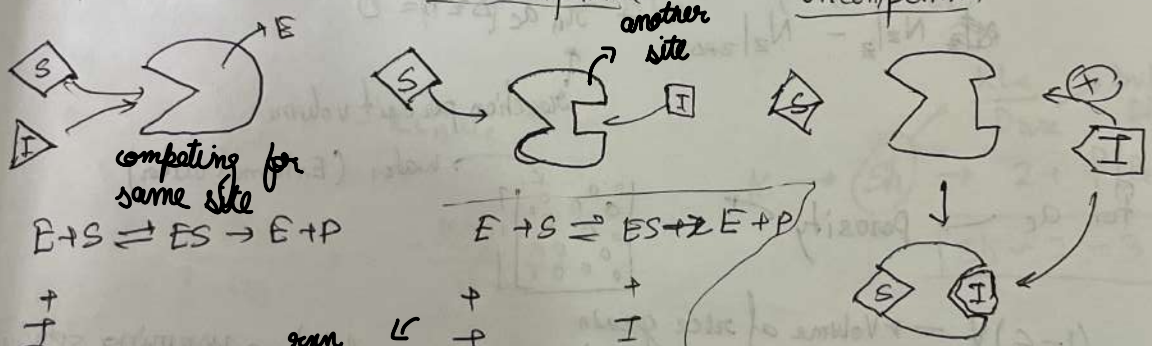
$$\Updownarrow \qquad\qquad \Updownarrow$$

$$EP \overset{+S}{\rightleftharpoons} ESP$$

(undesired) Not go to product (decreasing produces) (undesired)

(*) Inhibition → triggers additional reaction pathways, delay & reduce product formation

Competitive | Non competitive | Uncompetitive → not very common



$E + S \rightleftharpoons ES \rightarrow E + P$

$+I$

$\Updownarrow$

EI

rxn pathway altered

$E + S \rightleftharpoons ES \rightarrow E + P$

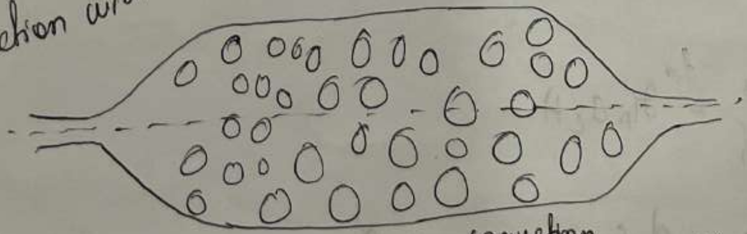
$+I \qquad +I$

$\Updownarrow \qquad \Updownarrow$

$EI + S \rightarrow ESI$

Packed-bed reactor

Reaction with Mass Transfer → convection, Diffusion



Reaction & diffusion & convection happens simultaneously.

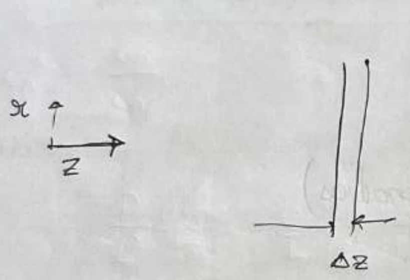
How do you model the performance / design scale-up of this system —

Possible Assumption

1. Isothermal operation
2. steady-state
3. No fluidization
4. No radial transport
5. Incompressible fluid & flow
6. Axial convection is dominant over diffusion.
   ↳ can be ignored for liquid system, not for gas
7) Catalyst particles are of uniform shape and size
8) Velocity is plug type

Compressible / Incompressible fluid → $\rho(P)' \neq 0$

Compressible / Incompressible flow → v_T speed of sound in the medium



$$N_A = \left[-D\frac{\partial C_A}{\partial z} + u_0 C_A\right] A$$

Mass conservation across Δz

$$N_z|_z - N_z|_{z+\Delta z} + r_A a_c (\Delta z A) = 0$$

a_c → specific surface area / Volume of reactor; effective catalyst surface area.

r_A → rate of reaction per unit volume.

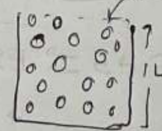
A → cross section area of packed bed

$$a_c = \frac{6(1-\phi_b)}{d_p}$$

ϕ_b → bed porosity

d_p → catalyst particle diameter

(*) For a_c → porosity



water (External water)

$(1-\epsilon)V$ → Volume of rice grain → we can get 'd' → assuming spherical → get surface area

$(1-\epsilon)V\rho_p$ → mass of rice grain

$$N_z = \left(-D\frac{dC_A}{dz} + vC_A\right)A$$

ignore → for liquid systems (unlike gases ↳ udhar diffusion bhi significant)

vC_A → plug flow velocity

$$\frac{dN_z}{dz} = r_A a_c A$$

↳ $$\cancel{A}\frac{d}{dz}(vC_A) = r_A a_c \cancel{A}$$

$$-\frac{\partial}{\partial z}(v C_A) + r_A a_c = 0$$

How do you describe flow ~~in~~ porous media?

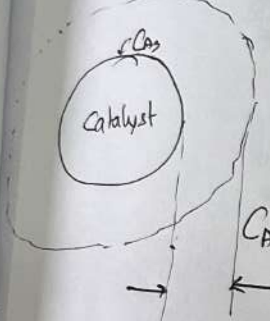
Darcy's Law $\longrightarrow$ $v = -\frac{K}{\mu}\nabla p$ ← permeability

$$v = -\frac{K}{\mu}\frac{dp}{dz} \rightarrow \text{constant}$$

v = constant

$$v\frac{dC_A}{dz} = r_A a_c$$

rxn @SS, so molar flux of species A on catalyst particle surface is equal to rate of disappearance of A from surface → catalytic rxn



C_A (bulk conc.)

surface conc. is negligible compared to bulk

At S.S

- rate at arriving at catalyst = rate of depletion

Both flux should be equal

$C_A >> C_{As}$

$K = \frac{D}{\delta}$ ← MTC

$$-r_p = K(C_A - C_{As})$$

rate of disappearance; K → MTC; $(C_A - C_{As})$ → Inward flux

Here we are assuming that for mass transport limited reactions, whatever is coming to particle surface is immediately consumed by reaction so $C_A >> C_{As}$

$$-r_p = K C_A \quad (C_A >> C_{As})$$

$$v\frac{dC_A}{dz} = -K C_A a_c$$

$K \rightarrow Sh \rightarrow 2 + f(Re, Sc)$; $Sh = \frac{K L_c}{D_{bulk}}$ → bulk Diffusion

estimation of K

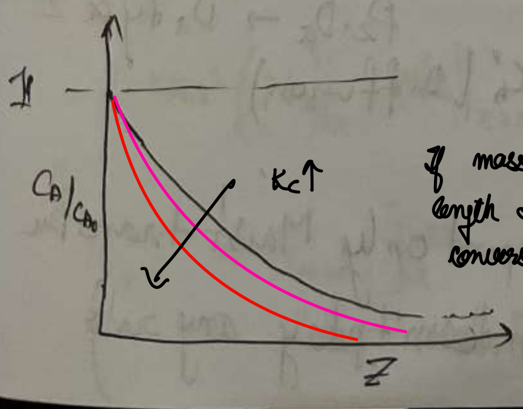
$\boxed{Sh \sim 2-3}$

$$\int\frac{dC_A}{C_A} = -\frac{K a_c}{v}\int dz$$

$$\ln\left(\frac{C_A}{C_{A_0}}\right) = -\frac{K a_c}{v} z$$

at $z=0$, $C_A = C_{A_0}$

$$C_A = C_{A_0}\exp\left(-\frac{K a_c}{v}z\right) \rightarrow C_{A_0}\exp\left(-\frac{K a_c L}{v}z^*\right)$$



If mass transfer coefficient is large then, smaller length of bed will be required to achieve same conversion

Relaxing assumption 6 ⟶ Both convec & Diffusion.

(Imp for gases)

$$-\frac{\partial}{\partial z}\left(-D\frac{\partial C_A}{\partial z} + vC_A\right) + r_A a_c = 0$$

$$D\frac{\partial^2 C_A}{\partial z^2} - v\frac{\partial C_A}{\partial z} + r_A a_c = 0$$

$$D\frac{\partial^2 C_A}{\partial z^2} - v\frac{dC_A}{dz} - kC_A a_c = 0$$

Non-Dimensionalise -

$$C_A^* = \frac{C_A}{C_{A_0}} \; ; \; z^* = z/L$$

$$Pe = \frac{vL}{D_A}$$

$$Da = \frac{k_c a_c L}{v}$$

$$\frac{D}{L^2}\frac{d^2C_A^*}{dz^{*2}} - \frac{v}{L}\frac{dC_A^*}{dz^*} - kC_A^* a_c = 0$$

$$\cancel{\frac{d^2C_A^*}{dz^{*2}}} - \cancel{\frac{vL}{D}}\frac{dC_A^*}{dz} - \cancel{\frac{kL^2 a_c}{D}C_A^*} = 0 \qquad \frac{D}{vL}\frac{d^2C_A^*}{dz^{*2}} - \frac{dC_A^*}{dz^*} - \frac{ka_cL}{v}C_A^* = 0$$

If $Pe \to$ large (convection is imp)

$\frac{1}{Pe} \to 0$, diffusive Term is not important

$$\frac{1}{Pe}\frac{d^2C_A^*}{dz^{*2}} - \frac{dC_A^*}{dz} - Da\, C_A^* = 0$$

$$Da = \frac{ka_cL}{v} = \text{type 1 } Da$$

BCs: $C_A^*(@\, z^*=0) = 1$

$$\left.\frac{dC_A^*}{dz^*}\right|_{@\, z^*=1} = 0$$

No concen change
No reaction ~~as conc~~ because No catalyst
No further reaction

(i) convection is smaller → then peclet No. is high, 1st term goes away → we get the same derived before.

(ii) $\left(Pe \sim \frac{1}{Da}\right) \ll 1 \Rightarrow \frac{1}{Pe}\frac{d^2C_A^*}{dz^{*2}} - D_A C_A^*$ (Diffusion)

$Pe \cdot Da \to$ 'Da type 2'

(iii) $Da \ll 1 \Rightarrow$ If ~~Reaction~~ transport depend only Mass transfer (Reaction doesn't play any role)

(iv) Da ≫ 1

How to estimate

Dispersion

Reach... throug... volum...

(iv) $Da \gg 1 \Rightarrow$ Reaction is dominant, transport is governed by reaction.

↳ So it doesn't bother, if we are in packed bed with catalyst as if we are outside with catalyst → both will be same

How to estimate C_{As} → If it is not neglegible

Dispersion

$$\frac{\partial c}{\partial t} + u\nabla C = D\nabla^2 c$$

γ (Dispersion Co-efficient) $= D_{porous} + U_0 L_c$ ↗ superficial velocity

$$\frac{\partial c}{\partial t} + u\nabla C = \gamma\nabla^2 c$$

$$\boxed{\frac{\partial c}{\partial t} = \gamma\nabla^2 c}$$

→ $D_{porous} \sim \phi D_{bulk}$

⊛ Catalyst particles are porous, ϕ_p

ϕ_p is large

Reaction is happening through out the volume

Homogenously through the particle → Reaction is considered to be happen almost everywhere inside the particle

ϕ_p is small

Reaction is happens only at pore

Not happens throughout

porous

⊛ Reaction happen only on pore surfaces

⊛ Not in the solid

ϕ_p is large → r, θ, ϕ $\quad \theta, \phi$ → symmetric

$R_A = kC_{Ar}$

C_{Ar} → conc. inside particle

$$\left(4\pi r^2 \{N_r\}\big|_r - (4\pi r^2 N_r)\big|_{r+\Delta r}\right) + r_A\, 4\pi r^2 \Delta r = 0 \rightarrow -\frac{d}{dr}(r^2 N_A) - r^2 R_A = 0$$

$N_r = -D_{porous}\dfrac{\partial C_{Ar}}{\partial r}$ → fick's law of diffusion

↳ $\sim \phi_p D_{Bulk}$

$$\frac{D}{r^2}\frac{d}{dr}\left(r^2\frac{\partial C_{Ar}}{\partial r}\right) + r_A = 0$$

$r_A = -K_{rxn} C_{Ar}$

symmetry BC

@ $r = 0, \quad \dfrac{\partial C_{Ar}}{\partial r} = 0$

$$\frac{D_p}{r^2}\frac{d}{dr}\left(r^2\frac{\partial C_{Ar}}{\partial r}\right) - K_{rxn} C_{Ar} = 0$$

@ $r = R_p, \quad C_{Ar} = C_{A_S}(z) \longrightarrow$ (conc^n continuity)

coupling to packed bed reactor

@ $r = R_p, \quad N_A = K_{mass}\left(C_A(z) - C_{A_S}\big|_{r=R_p}\right) = -D_p\dfrac{\partial C_{Ar}}{\partial r}$ (flux continuity)

Particle radius

$C_{Ar}(r, z)$

Bed-coordinate

$$\frac{d}{dr}\left(r^2\frac{dC_{Ar}}{dr}\right) = \frac{K}{D_{AB}}\frac{C_A}{r^2}$$

$$r^2\frac{dC_A}{dr} = \frac{K_r}{D_{AB}}\frac{C_A}{r} + \alpha_1$$

$(C_{AS} \to C_{Ar}|_{r=R_p})$

$$\boxed{-r_A = K_{mass}\left(C_A - C_{Ar}\big|_{r=R_p}\right)}$$

Define $\theta = C_A/C_{AS}$

& $\eta = r/R$

$$\frac{1}{\eta^2}\frac{d}{d\eta}\left(\eta^2\frac{d\theta}{d\eta}\right) = m^2\theta$$

$$m^2 = \frac{KR^2}{D_{AB}}$$

@ $\eta = 0 \quad \partial\theta/\partial\eta = 0$

@ $\eta = 1 \quad \theta = 1$

$f(C_{Ar}) \longrightarrow$ Continuous

$f'(C_{Ar}) \longrightarrow$ not continuous $\longrightarrow$ we need to specify

$$\frac{D}{r^2}\frac{d}{dr}\left(r^2\frac{\partial C_{Ar}}{\partial r}\right)$$

$$\cancel{D\left[\frac{\partial^2 C_{Ar}}{\partial r^2} + \frac{1}{r}\frac{\partial C_{Ar}}{\partial r}\right] = K_r C_{Ar}}$$

$$\left(\frac{1}{r}\frac{du^*}{dr} - \frac{U^*}{r^2}\right)$$

$$\cancel{\frac{\partial^2 C_{Ar}}{\partial r^2} + \frac{2}{r}\frac{\partial C_{Ar}}{\partial r} - \frac{K_r}{D}C_{Ar} = 0}$$

$$-\frac{\partial}{\partial r}\left(r\frac{du}{dr} - U\right) \qquad \frac{\partial U}{\partial r} + r\frac{\partial^2 U}{\partial r^2} - \frac{\partial U}{\partial r}$$

$$C_{Ar} = \frac{C^*}{r}$$

$$\frac{D}{r^2}\frac{\partial}{\partial r}\left(-\frac{\partial C^*}{\partial r}\right) = K_r\frac{C^*}{r}$$

$$\frac{\partial C_A}{\partial r} = -\frac{C^*}{r^2} + \frac{dC^*}{dr}$$

$$\boxed{\frac{d^2u}{dr^2} = \frac{K_r}{D_{AB}}\left(\frac{C_{Ar}}{r}\right)} \quad u$$

$$r^2\frac{\partial C^*_{Ar}}{\partial r} = -\frac{dC^*}{\partial r^2}$$

$$\frac{\partial}{\partial r}\left(r^2\frac{\partial C^*_{Ar}}{\partial r}\right) =$$

$$\frac{\partial^2 C^*_A}{\partial r^2} =$$

D_p

$$C_{Ar} = \frac{u}{r}$$

Use $\theta = u/\eta$

$$\frac{d^2u}{d\eta^2} = m^2 u$$

$\eta = 1 \quad u = 1$

$\eta = 0 \quad d\theta/d\eta = 0$

$\eta\dfrac{du}{d\eta} - u = 0$

$u = 0$

$$\frac{\partial^2 u}{\partial r^2} = \left(\frac{K_r}{D_{AB}}\right)u$$

@ $r = R_p, \quad u = R_p C_{Ar_p}$

@ $r = 0, \quad u = 0$

$$\alpha = A \sinh\left(\sqrt{\frac{K_r}{D_{AB}}}\, r\right) + B \cosh\left(\sqrt{\frac{K_r}{D_{AB}}}\, r\right)$$

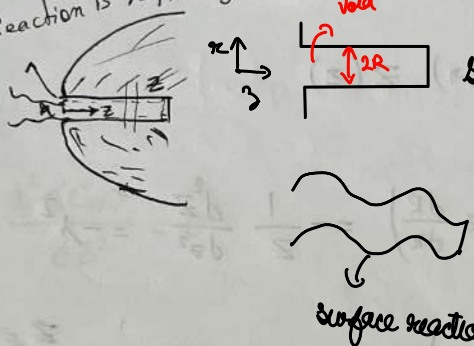
$\hookrightarrow A\sinh(m\eta) + B\cosh(m\eta)$

$$u = \frac{R_p C_{Arp} \sinh\left(\sqrt{\frac{K_r}{D_{AB}}}\, r\right)}{\cancel{R_p C_{Arp}} \sinh\left(\sqrt{\frac{K_r}{D_{AB}}}\, R_p\right)}$$

$$C_{Ar} = \frac{u}{r} = \frac{C_{Arp} R_p}{r} \frac{\sinh\left(\sqrt{K_r/D_{AB}}\, r\right)}{\sinh\left(\sqrt{K_r/D_{AB}}\, R_p\right)}$$

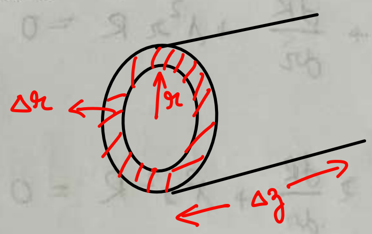
when porosity is low,

Reaction is happening at the wall, and rest is diffusion (only)

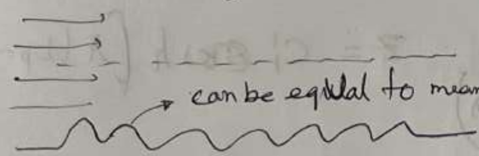


Mass conservation eq^n →

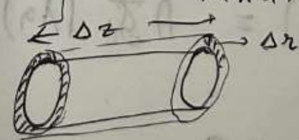
Approximation or assumption: Characteristic length scale is larger than mean free path length of fluid molecule → Continuum assumption is valid.



How do we define roughness



$$K_n = \frac{\text{molecular mean free path length}}{\text{characteristic length scale}} << 1$$ Continuum theory is valid.



mass conservation eq^n →

$$(2\pi r N_r)\big|_r - (2\pi r N_r)\big|_{r+\Delta r} + (2\pi \Delta r\, z N_z)\big|_z - (2\pi \Delta r\, z N_z)\big|_{z+\Delta z} = 0$$

$\frac{1}{r}\frac{\partial}{\partial r}(r N_r) - \frac{\partial}{\partial z}(N_z)$ $\quad N_{Ar} = -D_{AB}\frac{\partial C_A}{\partial r}$

$N_{Az} = -D_{AB}\frac{\partial C_A}{\partial z}$

$$\frac{D}{r}\frac{\partial}{\partial r}\left(r\frac{\partial C_A}{\partial r}\right) + D\frac{\partial^2 C_A}{\partial z^2} = 0$$

BC: $z = 0, \quad C_A = \overset{= C_{AS}}{C_A(\text{bulk})}$ $\quad r = 0, \quad \frac{\partial C_A}{\partial r} = 0$

$z = L_p, \quad \frac{\partial C_A}{\partial z} = 0 \quad (N_z = 0)$ $\quad r = R_p$ ↳ pore radius, $\quad D\frac{\partial C_B}{\partial z} = k_{rxn}C_B$

$r = 0 \quad \frac{\partial C_A}{\partial r} = 0$

Imp BCs ← $r = R \quad -D\frac{\partial C_A}{\partial r} = kC_A$ (flux)

$$C_A = R(r)\, Z(z)$$

$$\frac{1}{Rr}\frac{d}{dr}\left(r\frac{dR}{dr}\right) = -\frac{1}{Z}\frac{d^2 Z}{dz^2} = -\lambda^2$$

$$\frac{1}{Rr}\frac{d}{dr}\left(r\frac{dR}{dr}\right) + \lambda^2 = 0$$

$$r\frac{d^2R}{dr^2} + \frac{dR}{dr} + \lambda^2 r R = 0$$

$$r^2\frac{d^2R}{dr^2} + r\frac{dR}{dr} + \lambda^2 r^2 R = 0$$

$$Z = C_1 e^{+\lambda z} + C_2 e^{-\lambda z}$$

$$0 = C_1\cancel{\lambda} e^{\lambda L_p} + C_2\cancel{\lambda} e^{-\lambda L_p}$$

$$C_2 = C_1 e^{2\lambda L_p}$$

$$Z = 2C_1 e^{\lambda L_p}\left(\frac{e^{-\lambda(L_p - z)} + e^{\lambda(L_p - z)}}{2}\right)$$

$$Z = C_1' \cosh\left[\lambda(L_p - z)\right]$$

$$R(r) = A J_0(\lambda r) + B Y_0(\lambda r)$$

$r = 0, \quad R \to$ fin' $\quad \to B = 0$

$$R(r) = A J_0(\lambda r)$$

$$r = R_p \to D\frac{dR}{dr} - k_{rxn}R = 0$$

$$A D J_0'(\lambda r)$$

Area Averaging R_p

$$\langle C_A \rangle = \frac{\int_0^{R_p} 2\pi r\, C_A\, dr}{\pi R_p^2}$$

$$2\pi \int_0^{R_p} \frac{1}{r}\frac{\partial}{\partial r}\left(r \frac{\partial C_A}{\partial r}\right) r\, dr + 2\pi \int_0^{R_p} r \frac{\partial^2 C_A}{\partial z^2}\, dr = 0$$

$$2\left(r \frac{\partial C_A}{\partial r}\right)\Bigg|_{r=0}^{r=R_p} + \frac{d^2 \langle C_A \rangle}{dz^2}\frac{R_p^2}{2} = 0$$

$$R_p \cancel{\frac{\partial C_A}{\partial r}}\Bigg|_{R_p} \frac{K_{rxn} C_A|_{R_p}}{D}$$

$$\frac{2K}{R_p D} C_A|_{r=R_p} + \frac{d^2 \langle C_A \rangle}{dz^2} = 0$$

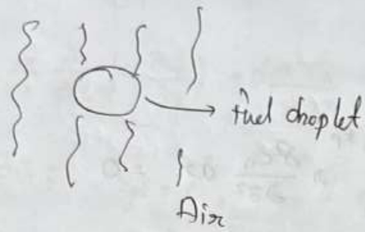
$$\langle C_A \rangle \sim C_A|_{r=r_p}$$

$$\frac{2K}{R_p D}\langle C_A \rangle + \frac{d^2 \langle C_A \rangle}{dz^2} = 0$$

BC @ $z=0$ $\quad \langle C_A \rangle = C_{AS}$

@ $z=L$ $\quad \frac{d\langle C_A \rangle}{dz} = 0$

Fuel Combustion



Mathematical model of the physical process of this fuel droplet combustion ?

Key processes :

1. Diffusion (Gas absorption)
2. Combustion / reaction
3. Heat transfer
4. Droplet would shrink &
5. Droplet itself can be non-stationary.

Assumptions:

Impractical

1. No shrinkage (droplet shape & size is time invariant)
2. Fix reference frame (moving) & to the object

common for spherical coordinates

3. θ & ϕ symmetry exists (droplet is spherical)
4. Non-heat transfer
5. No rxn (isothermal)

Impractical

6. Diffusivity is constant

8. Only applicable for

for diffusion model $\rightarrow$

Inward flux, outlet flux, Accumulation

$$\left(4\pi r^2 N_r\right)\Big|_r - \left(4\pi r^2 N_r\right)\Big|_{r+\Delta r} = \frac{\partial}{\partial t}\left(4\pi r^2 \Delta r\, C\right)$$

~~For~~ sdi.

$$4\pi r^2 \frac{\partial c}{\partial t} = -4\pi \frac{\partial}{\partial r}\left(r^2 N_r\right)$$

$N_r = -D\frac{\partial c}{\partial r}$ (fick law)

$$\cancel{\frac{\partial c}{\partial t} = \frac{1}{r^2}\frac{\partial}{\partial r}\left[r^2 \frac{\partial}{\partial r}\left(D\frac{\partial c}{\partial r}\right.\right.}$$

$$\frac{\partial c}{\partial t} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 D \frac{\partial c}{\partial r}\right)$$

Assumption →

(6) Only applicable for binary system, film should be stagnant, component should be ideal in nature, dilute conc^n

↳ Stefan maxwell eqn deals with conc^n system

$\boxed{P_{vap} = \rho RT}$

~~$P_{sat} = \rho RT$~~

$$\frac{\partial c}{\partial t} = \frac{D}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial c}{\partial r}\right) \rightarrow \text{governing equation}$$

@ $r=0$, $\frac{\partial c}{\partial r} = 0$ → symmetry

@ $r = R$, $C = C_\infty$ → (we can calculate from Henry law) → equilibrium solubility

@ $t = 0$, $C = 0$ (or C_0)

$$\tilde{C} = \frac{C - C_\infty}{C_0 - C_\infty} \qquad \tilde{r} = r/R$$

$$\tilde{t} = \frac{tD}{R^2} \qquad t^* = R^2/D \Rightarrow \text{diffusive time scale}$$

$$\frac{\partial \tilde{C}}{\partial \tilde{t}} = \frac{1}{\tilde{r}^2}\frac{\partial}{\partial \tilde{r}}\left(\tilde{r}^2\frac{\partial \tilde{C}}{\partial \tilde{r}}\right)$$

Let $u = \tilde{C}\tilde{r}$

@ $\tilde{r} = 0$, $\frac{\partial \tilde{C}}{\partial \tilde{r}} = 0$

$\frac{u}{\tilde{r}} = \tilde{C}$ } Imp transformation

@ $\tilde{r} = 1$, $\tilde{C} = 0$

@ $\tilde{t} = 0$, $\tilde{C} = 1$

$$\frac{\partial \tilde{C}}{\partial \tilde{t}} = \frac{1}{\tilde{r}}\frac{\partial u}{\partial \tilde{t}}$$

$$\frac{\partial \tilde{C}}{\partial \tilde{r}} = \frac{1}{\tilde{r}}\frac{\partial u}{\partial \tilde{r}} - \frac{u}{\tilde{r}^2}$$

$$\tilde{r}^2\frac{\partial \tilde{C}}{\partial \tilde{r}} = \tilde{r}\frac{\partial u}{\partial \tilde{r}} - \tilde{u}$$

$$\tilde{r}^2\frac{\partial \tilde{C}}{\partial \tilde{r}} = \frac{\tilde{r}^2}{\tilde{r}}\frac{\partial u}{\partial \tilde{r}} - \frac{u}{\tilde{r}^2}\tilde{r}^2$$

$$\frac{\partial}{\partial \tilde{r}}\left(\tilde{r}^2 \frac{\partial \tilde{c}}{\partial \tilde{r}}\right) = \frac{\partial \tilde{u}}{\partial \tilde{r}} + \tilde{r}\frac{\partial^2 u}{\partial \tilde{r}^2} - \frac{\partial \tilde{u}}{\partial \tilde{r}}$$

$$\frac{1}{\tilde{r}^2}\frac{\partial}{\partial \tilde{r}}\left(\tilde{r}^2 \frac{\partial \tilde{c}}{\partial \tilde{r}}\right) = \frac{1}{\tilde{r}}\frac{\partial^2 u}{\partial \tilde{r}^2}$$

$$\therefore \frac{\partial^2 u}{\partial \tilde{r}^2} = \frac{\partial u}{\partial \tilde{t}}$$

@ $\tilde{r} = 0$, $\cancel{\frac{\partial u}{\partial \tilde{r}} \neq 0}$, $\cancel{\tilde{r}^2}\frac{\partial \tilde{c}}{\partial \tilde{r}} = \cancel{\tilde{r}}\frac{\partial u}{\partial \tilde{r}} - u = 0 \Rightarrow u = 0$

@ $\tilde{r} = 1$, $u = 0 \Rightarrow v = \tilde{c}\,\tilde{r} = 0$

@ $\tilde{t} = 0$, $u = \tilde{c}\,\tilde{r} \Rightarrow u = \tilde{r}$

$$u = G(\tilde{r})\,F(\tilde{t})$$

$$\frac{1}{G}\frac{d^2 G}{d\tilde{r}^2} = +\frac{1}{F}\frac{dF}{d\tilde{t}} = -\lambda^2$$

$$G = C_1 \sin(\lambda_m \tilde{r})$$

$$F = A \exp(-\lambda_m^2 t)$$

$$\therefore u = \sum_{m=1}^{\infty} C_m \exp(-\lambda_m^2 t)\sin(\lambda_m \tilde{r})$$

$\lambda_m = m\pi$

$$\int_0^1 \tilde{r}\sin(\lambda_m \tilde{r})\,d\tilde{r} = C_m\int_0^1 \sin^2(\lambda_m \tilde{r})\,d\tilde{r} \qquad m = 1, 2, \ldots$$

$$C_m = \frac{\int_0^1 \tilde{r}\sin(m\pi\tilde{r})\,d\tilde{r}}{\int_0^1 \sin^2(m\pi\tilde{r})\,d\tilde{r}}$$

$$\therefore \quad \bar{C} = \frac{1}{\bar{r}} \sum_{m=1}^{\infty} C_m \exp(-m^2\pi^2 \bar{t}) \sin(m\pi \bar{r})$$

$$\frac{C - C_\infty}{C_0 - C_\infty} = \bar{C} \qquad \frac{C}{C_0 - C_\infty} = \bar{C} + \frac{C_\infty}{C_0 - C_\infty}$$

(molar) Rate of gas absorption,

$$\dot{m} = -4\pi r^2 N_r (r = R) \qquad C_A = K \int_0^t \phi(\eta, r) e^{-K\eta} d\eta + \phi(t, r) e^{-Kt}$$

$$= +4\pi r^2 D \left.\frac{\partial C}{\partial r}\right|_{r=R}$$

Now, if we relax the no rxn assumption.

$$\therefore \quad \frac{D}{r^2} \frac{\partial}{\partial r}\left(r^2 \frac{\partial C}{\partial r}\right) - KC = \frac{\partial C}{\partial t}$$

$$\frac{D}{R^2 \bar{r}^2} \frac{\partial}{\partial \bar{r}}\left(\bar{r}^2 \frac{\partial \bar{C}}{\partial \bar{r}}\right) - K\left(\bar{C} + \frac{C_\infty}{C_0 - C_\infty}\right) = \frac{\partial \bar{C}}{\partial t}$$

$$\frac{1}{\bar{r}^2} \frac{\partial}{\partial \bar{r}}\left(\bar{r}^2 \frac{\partial \bar{C}}{\partial \bar{r}}\right) - \frac{KR^2}{D}\left(\bar{C} + \frac{C_\infty}{C_0 - C_\infty}\right) = \frac{\partial \bar{C}}{\partial \bar{t}}$$

Danckwerts solution $\Rightarrow$ (1968)

$$C = K \int \phi e^{-K\bar{r}} d\bar{r} + \phi e^{-Kt} \qquad \phi(\eta, r) \quad \phi(t, r)$$

where $\phi \equiv$ is the solution without source term

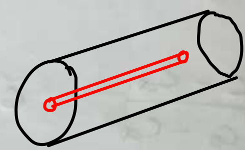
Now, relaxing the No shrinkage Assumption:

shrinkage effect $\Rightarrow$ @ $r = R(t)$; $C = C_\infty$ $\rightarrow \frac{dR}{dt}$

$\Downarrow$

There will be pseudo convective term appear $\Leftarrow$ moving boundary effect in the balance eqn

Oxygenation of tissues

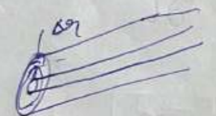


- Modeling the species (mostly O_2) profile in the tissue region

Assumption:

1) Radial diffusion   Steady flow, unsteady O_2 transfer
2) No chemical Reaction → b/w O_2 & Tissue
3) Isothermal
4) Diffusivity of O_2 is const

Mathematical eqⁿ



$$\Delta V \frac{dy}{dt} = N_{O_2}(2\pi r Z)\Big|_r - N_{O_2} 2\pi(r+\Delta r)Z\Big|_{r+\Delta r} - \Delta V \alpha_{O_2}$$

$$\Delta V = \pi\left[(r+\Delta r)^2 - r^2\right]Z$$

$$\cancel{Z}(2\pi r N_r)\Big|_r - \cancel{Z}(2\pi r N_r)\Big|_{r+\Delta r} = \frac{\partial}{\partial t}(2\pi r \Delta r \cancel{Z} C)$$

α_{O_2}: rate of O_2 absorption in tissue

$$-\frac{1}{r}\frac{\partial}{\partial r}(r N_r) = \frac{\partial c}{\partial t}$$

IC $y(r,0) = y_0$ $R_1 < r \le R_2$

BC @ $r = R_1$ $y = y_F$

@ $y = R_2$ $\partial y / \partial r = 0$

$$N_r = -D\frac{\partial c}{\partial r}$$

$$\therefore \frac{D}{r}\frac{\partial}{\partial r}\left(r\frac{\partial c}{\partial r}\right) = \frac{\partial c}{\partial t}$$

from henry' law → $c \sim (\text{const})\, p_{O_2}$

$$\frac{\partial p_{O_2}}{\partial t} = \frac{D}{r}\frac{\partial}{\partial r}\left(r\frac{\partial p_{O_2}}{\partial r}\right)$$

BC:

@ $r = R_1$ (radius of the capillary / blood vessel)

$$p_{O_2} = p_{O_2}^*$$

@ $r \to \infty$, $\frac{\partial p_{O_2}}{\partial r} \to 0$ $(N_r \to 0)$

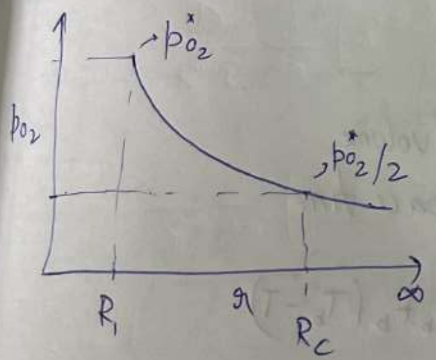
@ $t = 0$, $p_{O_2} = 0$ or $p_{O_2} = p_{O_2}^o$ $(<< p_{O_2}^*)$

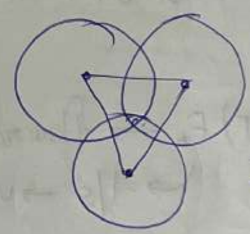
Now, if no chemical rxn assumption is relaxed,

Chemical rxn →

$$\frac{\partial C}{\partial t} = \frac{D}{r}\frac{\partial}{\partial r}\left(r\frac{\partial C}{\partial r}\right) - \alpha C \quad \text{(cont)}$$

$$\frac{\partial p_{O_2}}{\partial t} = \frac{D}{r}\frac{\partial}{\partial r}\left(r\frac{\partial p_{O_2}}{\partial r}\right) - \bar{\alpha}\, p_{O_2}$$





In order to get all the area covered, ~~all the~~ their must be overlap

→ Golden Ratio

↓ Packing fraction

Heat transfer in tissue

Model the heat transfer process in the tissue.

Assumption: (1) Radial conduction

(2) Net Homogenous source term

$$\cancel{\rho C_p \frac{\partial T}{\partial t} = k\frac{\partial^2 T}{\partial r^2} + \dot{q}}$$

Mathematical model

$$\cancel{z(2\pi r q_r)|_r - z(2\pi r q_r)_{r+\Delta r} = \frac{\partial}{\partial t}}$$

$$z(2\pi r q_r)|_r - z(2\pi r q_r)|_{r+\Delta r} + \underbrace{\dot{q}\, 2\pi r \Delta r z}_{\dot{m} C_{p_b} \Delta T} = \frac{\partial}{\partial t}(\rho C_p 2\pi r \Delta r z T)$$

$$-\frac{1}{r}\frac{\partial}{\partial r}(r q_r) + \dot{q} = \rho C_p \frac{\partial T}{\partial t}$$

$$q_r = -k\frac{\partial T}{\partial r}$$

$$\rho C_p \frac{\partial T}{\partial t} = \frac{k}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T}{\partial r}\right) + \dot{q}$$

$\dot{q} = \rho_b C_{p_b}(T_b - T)F_b \rightarrow$ flow rate blood/volume

$\hookrightarrow 1/s \rightarrow$ unit $\rightarrow$ space time

$$\therefore \rho C_p \frac{\partial T}{\partial t} = \frac{k}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T}{\partial r}\right) + \rho_b C_{p_b} F_b (T_b - T)$$

$\rho_b, C_{p_b}, T_b, F_b \rightarrow$ for blood

$\rho, C_p, T, K \rightarrow$ for tissue

BCs: @ $r = 0$, $\frac{\partial T}{\partial r} = 0$

@ $r = R$, $\boxed{q_r = -k\frac{\partial T}{\partial r}\Big|_{r=R} = h(T - T_\infty)}$

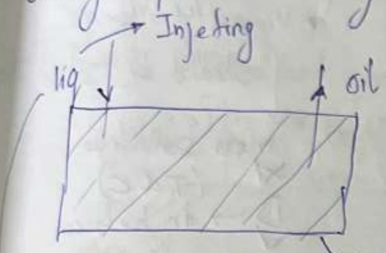
$R \downarrow$ outer surface

IC: @ $t = 0$, $T = T_0$

Oil recovery Problem

[solvent assisted Gravity drainage] — SAGD

⊛ Due to high viscosity, the pressure drop is very high or very high suction pressure. (we cannot apply very pressure, because crude oil may vapourise leading to ~~b~~ (blockage)

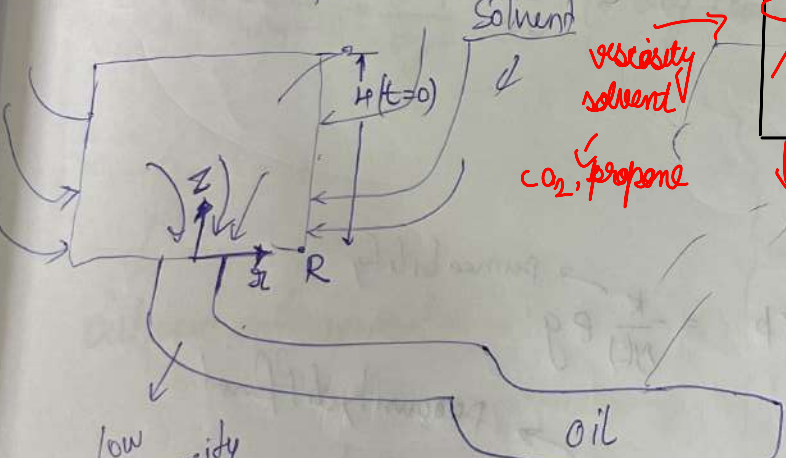


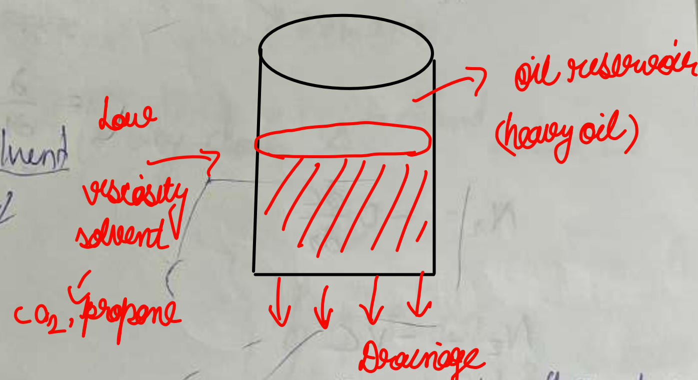
classical oil recovery process

- low viscosity
- easily seperatable {Water, steam, CO_2, CH_4} → Solvent

also known as

Modern Approach (SAGD)





This can be other mines

⊛ model (i) the diffusion of solvent in this (depleted) oil reservoir ??

(ii) Oil production rate ?

Assumption

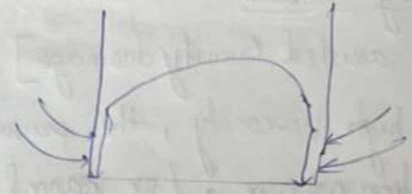
(i) Cylindrical domain — Solvent penetrates uniformly from all sides of reservoir

(ii) oil level shrinks along z-direction (due to gravity)

(iii) Isothermal

(iv) No chemical reaction

(v) ρ is unaffected



(5% to 10% is introduced)

↳ Viscosity change

Should be short

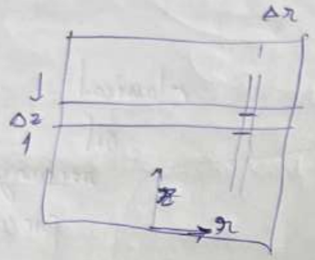
↓ more sensitive

μ (T & C)

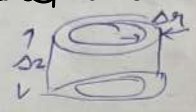
D → in between

ρ (density)

↳ least change (T & C)



Diffusion along axial dirn small compared to convection



$$2\pi r \Delta z N_r\big|_r - 2\pi(r+\Delta r)\Delta z N_r\big|_{r+\Delta r} + 2\pi r \Delta r N_z - 2\pi r \Delta r N_z\big|_{z+\Delta z}$$

$$= \frac{\partial}{\partial t}\left(2\pi r \Delta r \Delta z\, c\, \phi\right)$$

$$N_r = -D\frac{\partial c}{\partial r} \quad \text{(No convection in r)}$$

$$N_z = -vC \quad \text{(No diffusion in z-direction)}$$

$$v = -\frac{K}{\mu}\nabla p = -\frac{K}{\mu(c)}\rho g$$

K → permeability

$$\phi\frac{\partial c}{\partial t} - \frac{\partial}{\partial z}\left(\frac{K}{\mu(c)}\rho g c\right) = \frac{D}{r}\frac{\partial}{\partial r}\left(r\frac{\partial c}{\partial r}\right)$$

D → porosity diffusivity

$$\phi\frac{\partial c}{\partial t} - K\rho g\frac{\partial}{\partial z}\left(\frac{c}{\mu(c)}\right) = \frac{D}{r}\frac{\partial}{\partial r}\left(r\frac{\partial c}{\partial r}\right)$$

$$\Delta V = 2\pi r \Delta r \Delta z$$

$$A_r = 2\pi r \Delta z$$

$$A_z = 2\pi r \Delta r$$

BC: @ $r=0$, $\dfrac{\partial c}{\partial r} = 0$

@ $r = R$, $c = c_s$

@ $z = H(t)$, $c = 0$

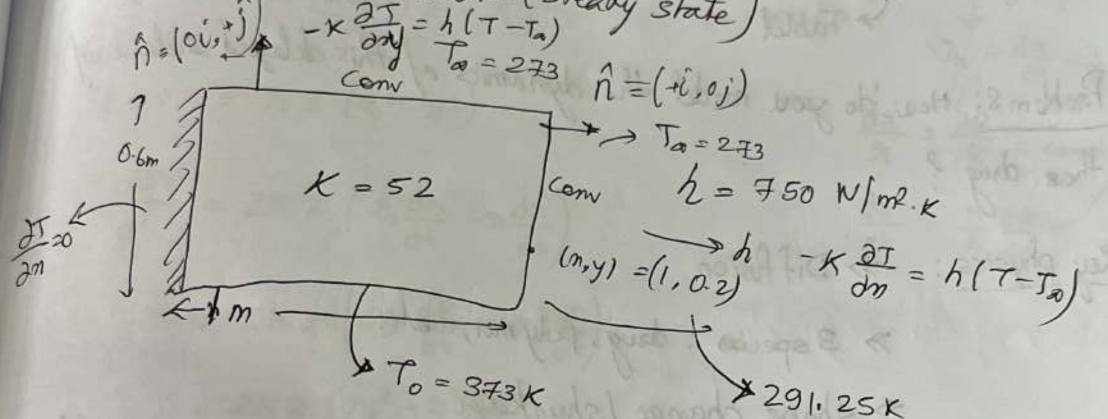
IC: @ $t = 0, \quad c = 0$

$$\frac{\partial}{\partial t}\left(\phi \rho \pi R^2 H \bar{c}\right) = -\int_0^R \rho \phi 2\pi r\, v\, dr \longrightarrow \text{conservation}$$

$\bar{c}$ → Average Volumetric conc^n

$v \rightarrow v(z=0)$

Problem: 2D Heat conduction (steady state)



$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0$$

$-n \cdot q = d$

↓

outward unit normal

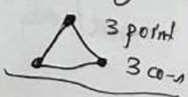
* why choose △ element instead □

⇓

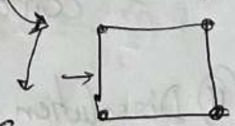
$\oint_\Omega \phi \ T \, d\Omega$

Simple form

$\alpha_0 + \alpha_1 x + \alpha_2 y$



D.O.F = 0

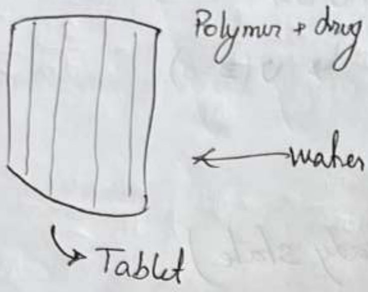


$\alpha_0 + \alpha_1 x + \alpha_2 y + \alpha_3 xy$

→ 4 co-effi → test function

? computationally heavy for quad

Controlled/Sustained release problem



Problems: How do you model the dynamics of this delayed release of the drug?

Key physics: ≫ Diffusion

≫ 3 species: drug, polymer, water

≫ shape change/shrinkage

≫ Varying density ⓘ Because water will be diffusing in, which will change density

Assumption: ① Isothermal

② Fick's law

③ shape change is dominant along r-direction.

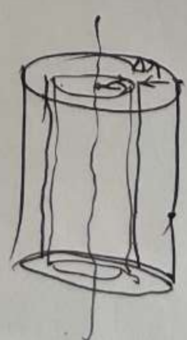
constant strain shrinkage $\frac{\Delta r}{r} \sim$ const

④ Dissolution rate of drug is very fast

↳ Diffusion of polymer is RTB

Mass conservation

$$N_{r_i} A_r|_r - N_{r_i} A_r|_{r+\Delta r} = \frac{\partial}{\partial t}(\rho_i \Delta V)$$



release of

diffusing change

RTB

$$-\frac{\partial}{\partial r}(N_r A_r) = \frac{\partial}{\partial t}(\rho_i \Delta V) \rightarrow \rho_i \underline{\frac{\partial \Delta V}{\partial t}} + \Delta V \frac{\partial \rho_i}{\partial t}$$

$$N_{ri} = \rho_i v_i - D_i \frac{\partial \rho_i}{\partial r} = \rho_i v_{ri} - D\rho \frac{\partial w_i}{\partial r}$$

Velocity → Because of moving Boundary

↓ Because there no convection

$$v_r = \frac{dr}{dt} = \frac{r}{R}\frac{dR}{dt}$$

$$\underset{\text{diff volume}}{\Delta V} = \pi \underset{r_s}{(r+\Delta r)}^2 z - \pi r^2 z$$

$$\Delta V = \pi z (r_s^2 - r^2)$$

$$\frac{dr}{r} = \frac{dR}{R} = \text{const}$$

$$\frac{dr}{dt} = \frac{r}{R}\frac{dR}{dt}$$

$$\frac{\partial(\Delta V)}{\partial t} = 2\pi z \left(r_s \frac{dr_s}{dt} - r\frac{dr}{dt}\right)$$

$$= 2\pi z \left(r_s^2 \frac{dR}{dt} - r^2 \frac{dR}{dt}\right)$$

$$= \frac{2\pi z (r_s^2 - r^2)}{R}\frac{dR}{dt}$$

$$= \frac{4\pi r \Delta r z}{R}\frac{dR}{dt}$$

we know, $A_r = 2\pi r z$

$$\Rightarrow 2\pi r z \left(\rho_i v_r - D_i \rho \frac{\partial w_i}{\partial r}\right)\Big|_r - 2\pi (r+\Delta r) z \left(\rho_i v_r - D_i \rho \frac{\partial w_i}{\partial r}\right)\Big|_{r+\Delta r}$$

$$= \rho_i \frac{4\pi r \Delta r z}{R}\frac{\partial R}{\partial t} + 2\pi r \Delta r z \frac{\partial \rho_i}{\partial t}$$

$$-\cancel{2\pi z}\frac{1}{r}\frac{\partial}{\partial r}\left(\rho_i v_r r - r D_i \rho \frac{\partial w_i}{\partial r}\right) = \rho_i \frac{\cancel{4\pi z}\,2}{R}\frac{dR}{dt} + \frac{\partial \rho_i}{\partial t}$$

$$-\frac{1}{r}\frac{\partial}{\partial r}(\rho_i v_r r) + \frac{D_i \rho}{r}\frac{\partial}{\partial r}\left(r\frac{\partial w_i}{\partial r}\right) = \frac{2\rho_i}{R}\frac{dR}{dt} + \frac{\partial \rho_i}{\partial t}$$

$$-\frac{1}{\cancel{r}}\rho_i \cancel{r}\frac{\partial}{\partial r}\left(\frac{r}{R}\frac{dR}{dt}\right) - \frac{1}{r} v_r \frac{\partial}{\partial r}(r \rho w_i) + \cdots$$

$$\Rightarrow -\rho_i \frac{\partial}{\partial r}\left(\frac{r}{R}\frac{dR}{dt}\right) - \frac{1}{\cancel{r}}\cdot\frac{\cancel{r}}{R}\frac{dR}{dt}\rho\frac{\partial}{\partial r}(r\rho_i) +$$

$$\Rightarrow \quad -2\frac{\rho_i}{R}\frac{dR}{dt} \quad - \frac{r}{R}\frac{dR}{dt}\frac{\partial \rho_i}{\partial r} - \cancel{\frac{\rho_i}{R}\frac{dR}{dt}}$$

$$\frac{d\rho_i}{dt} = -\frac{1}{R}\frac{dR}{dt}\left(4\rho_i + r\frac{\partial \rho_i}{\partial r}\right) + \frac{1}{r}\frac{\partial}{\partial r}\left(r D_i \frac{\partial \rho_i}{\partial t}\right)$$

$\rho_i(r,t)$

$w_i(r,t)$

$R(t)$

Balance of the polymer:

$$\frac{\partial}{\partial t}(\text{Polym in tablet}) = \text{Polymn las (dissolved) from surface.}$$

$$\frac{d}{dt}\int_0^R 2\pi r z \rho_2\, dr = -(2\pi R z) K_2 \left(\rho_2|_{r=R} - \cancel{\rho_2|_{r=\infty}}\right) \quad 0$$

$$\boxed{\frac{d}{dt}\int_0^R r\rho_2\, dr = -R K_2 \tilde{\rho}_2}$$

$$\cancel{\frac{dR}{dt} R\tilde{\rho}_2} - \cancel{Q} = \cancel{-R K_2 \tilde{\rho}_2}$$

$$\frac{dR}{dt}\ \frac{1}{R}\int_0^R r\rho_2\, dr$$

$$\# \qquad \frac{dR}{dt} = \frac{K\tilde{\rho}_2 R + \int_0^R \frac{\partial}{\partial r}\left(r D_2 \frac{\partial \rho_2}{\partial r}\right) dr}{\frac{1}{R}\int_0^R r\rho_2\, dr}$$

BC @ $r=0$, $\frac{\partial \rho_i}{\partial r}=0$

@ $r=R$, $\rho_1 \to 0$, $-D_2 \frac{\partial \rho_2}{\partial r} = \kappa \rho_2$; $-D_3 \frac{\partial \rho_3}{\partial r} = \kappa(\rho_3 - \rho_{\infty})$

IC @ $t=0$, $\rho_1 = \rho_{10}$, $\rho_2 = \rho_{20}$, $\rho_3 = 0$

Approximate Solution

Perturbation Methods

— Nayfeh (Author book)

Function : $\rightarrow u(x, \varepsilon)$

operator $\leftarrow L(u) = 0$] Equation

$u \rightarrow$ dependent variable

$x \rightarrow$ independent variable

$\varepsilon \rightarrow$ parameter

Solution $L(u) = 0$, for $\varepsilon = \varepsilon_0$ (where $\varepsilon_0 \to 0$)

↳ Approximate solution for small value of ε, around $\varepsilon_0 (\to 0)$

$$u(x;\varepsilon) = u_0(x) + \varepsilon u_1(x) + \varepsilon^2 u_2(x) + \varepsilon^3 u_3(x) + \cdots$$

$$= \sum_{n=0}^{\infty} \varepsilon^n u_n(x)$$

Example

$u = 1 + \varepsilon u^3 \rightarrow$ cubic eq^n (Van der-Pol oscillator)

consider ε to be small

Base case : $\varepsilon = 0$, $u_0 = 1$

$$u = u_0 + \varepsilon u_1 + \varepsilon^2 u_2 + \cdots$$

$$(u_0 + \varepsilon u_1 + \varepsilon^2 u_2 + \cdots) = 1 + \varepsilon (\overset{1}{u_0} + \varepsilon u_1 + \varepsilon^2 u_2 + \cdots)^3$$

$$\cancel{1} + \varepsilon u_1 + \varepsilon^2 u_2 + \cdots = \cancel{1} + \varepsilon(1 + \varepsilon u_1 + \varepsilon^2 u_2 + \cdots)^3$$

$$= \varepsilon(1^3 + 3\varepsilon u_1 1^2 + \cdots)$$

comparing co-efficient ε

$$u_1 = u_0^3 = 1$$

co-efficient of ε^2

$$u_2 = 3u_1 u_0^2 = 3$$

$$u = 1 + \varepsilon + 3\varepsilon^2 + \cdots$$

$$(a+b+c)^3 = (a+b)^3 + 3(a+b)^2 c + 3(a+b)c^2 + c^3$$

$$= a^3 + b^3 + 3a^2 b + 3ab^2 +$$

$$\frac{d^2u}{dt^2} + u = \varepsilon(1-u^2)\frac{du}{dt} \Rightarrow u = u_0 + \varepsilon u_1 + \varepsilon^2 u_2 + \varepsilon^3 u_3 + \cdots$$

Series solution in terms of (small) ε

Zeroth order solution : $\varepsilon \to 0$ (Leading)

$$\frac{d^2u_0}{dt^2} + u_0 = 0 \quad \text{(SHM)}$$

$$u_0 = a\cos(t + \phi)$$

first order solution, ε

$$\left(\frac{d^2u_0}{dt} + \varepsilon\frac{d^2u_1}{dt^2} + \cdots\right) + (u_0 + \varepsilon u_1) = \varepsilon\left(1 - (u_0 + \varepsilon u_1)^2\right)\left(\frac{du_0}{dt} + \varepsilon\frac{du_1}{dt}\right)$$

co-efficient 'ε^1'

$$\frac{d^2u_1}{dt^2} + u_1 = (\varepsilon^1 - \varepsilon^1 u_0)\frac{du_0}{dt}$$

$$\frac{d^2u_1}{dt^2} + u_1 = (1-u_0^2)\frac{du_0}{dt} \Rightarrow O(\varepsilon)$$

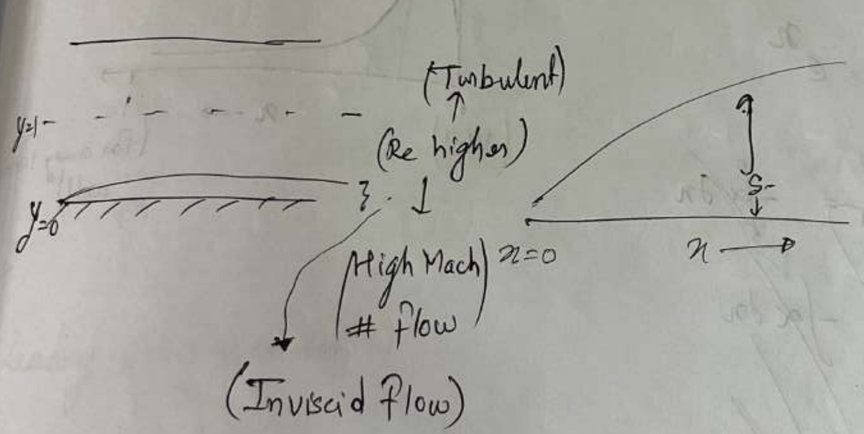
$$\frac{d^2u_1}{dt^2} + u_1 = -(1-a^2\cos^2(t+\phi))\, a\sin(t+\phi)$$

$$\boxed{\frac{d^2u_1}{dt^2} + u_1 = -a\sin(t+\phi)\left(1-a^2\cos^2(t+\phi)\right)}$$

$$O(\varepsilon^2) \Rightarrow \frac{d^2u_2}{dt^2} + u_2 = (1-u_0^2)\frac{du_1}{dt} - 2u_0u_1\frac{du_0}{dt}$$

$$\therefore\ u = u_0 + \varepsilon u_1 + \varepsilon^2 u_2 + \cdots$$

Boundary layers (Prantl Solution)



For flat plate

$$\delta \sim \frac{1}{Re^{1/2}}$$

Simple Boundary layer problem

$\varepsilon y'' + y' + y = 0$ (Prantl technique Turbulent BL)

BC: @ $y(0) = \alpha$ (could be zero)

$y(1) = \beta$ (center line velocity)

$\varepsilon \ll 1$

$$\frac{dy_0}{dx} = -y_0$$

Zeroth order sol^n:

(outer solution)

$\varepsilon \to 0$, $y_0' + y_0 = 0$ BC @ $y_0(1) = \beta$

$$\ln y_0 = -x + c$$

$$y_0 = Ae^{-x}$$

$y_0 = \cancel{\beta} e^{1-x}$

$y = y_0 + \epsilon y_1 + \epsilon y_2 + \cdots$

$\epsilon (y_0'' + \epsilon y_1'' + \epsilon^2 y_2'' + \cdots) + (y_0' + \epsilon y_1' + \epsilon^2 y_2' +) + (y_0 + \epsilon y_1 + \epsilon^2 y_2)$
$= 0$

comparing $\epsilon \rightarrow$ coefficient :

$y_0'' + y_1' + y_1 = 0$

$y_1' + y_1 + \cancel{\beta} e^{1-x} = \cancel{0}$

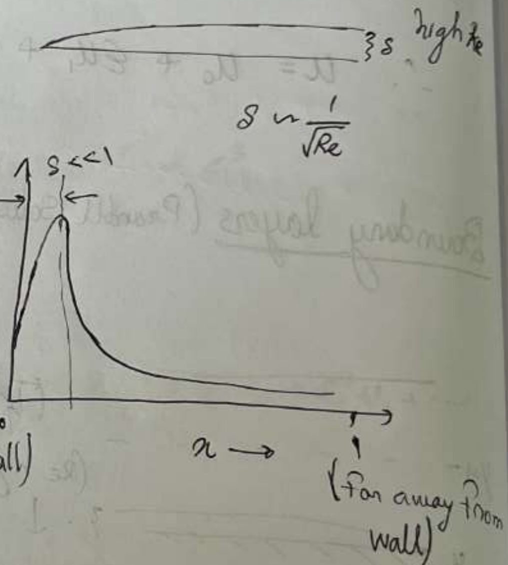
$y_1' + y_1 = -\cancel{\beta} e^{1-x}$

$I.F = e^{\int 1 dx} = e^{x}$

$e^{x} y_1' + e^{x} y_1 = -\cancel{\alpha} dx$

$\int d(e^{x} y_1) = -\int \alpha \, dx$

$e^{x} y_1 = -\alpha x + c_1$



Exact solution

$(\epsilon D^2 + D + 1) y = 0$

$S_{1,2} = \frac{-1 \pm \sqrt{1 - 4\epsilon}}{2\epsilon}$

$y = c_1 e^{S_1 x} + c_2 e^{S_2 x}$

at $y(x=0) = \alpha$

$$c_1 + c_2 = \alpha$$

$$c_1 e^{s_1} + c_2 e^{s_2} = \beta$$

$$c_1 + c_2 e^{s_2 - s_1} = \beta e^{-s_1}$$

$$c_2 (1 - e^{s_2 - s_1}) = (\alpha - \beta e^{-s_1})$$

$$c_2 = \frac{\alpha - \beta e^{-s_1}}{1 - e^{s_2 - s_1}} = \frac{e^{s_1}\alpha - \beta}{e^{s_1} - e^{s_2}}$$

$$c_1 = \frac{\alpha e^{s_1} - \alpha e^{s_2} - e^{s_1}\alpha + \beta}{e^{s_1} - e^{s_2}}$$

$$c_1 = \frac{\beta - \alpha e^{s_2}}{e^{s_1} - e^{s_2}}$$

$$\therefore \quad y = \frac{(\alpha e^{s_2} - \beta) e^{s_1 x} + (\beta - e^{s_1}\alpha) e^{s_2 x}}{e^{s_2} - e^{s_1}}$$

#

Leading order solution,

$$\varepsilon = 0$$

$$y' + y = 0 \qquad [\text{BC } y(x=1) = \beta]$$

outer solution $y_0 = \beta e^{1-x}$

→ Take care of wall effect.

Inner solution: magnify the scale → $\xi = \frac{x}{\varepsilon}$ (Not applicable when $x \to 1$)

$$\varepsilon \frac{d^2 y}{dx^2} + \frac{dy}{dx} + y = 0 \Rightarrow \frac{d^2 y}{d\xi^2} + \frac{dy}{d\xi} + \varepsilon y = 0$$

Leading order solution ($\varepsilon = 0$)

$$\frac{d^2y}{d\xi^2} + \frac{dy}{d\xi} = 0 \longrightarrow y = A + Be^{-\xi}$$

$y(x=0) = \alpha$ | $y = A + (\alpha - A)e^{-\xi}$

$y(\xi = 0) = \alpha$

$\alpha = A + B$

$B = \alpha - A$

$\therefore \lim_{\xi \to \infty}$ inner solution $= \lim_{x \to 0}$ outer solution

$$A = \beta e$$

$$\therefore y = \beta e + (\alpha - \beta e)e^{-x/\xi} \text{] Inner solution}$$

$$y = \beta e^{1-x} \text{] outer solution}$$

Complete solution:

Inner solution + Outer solution − (Inner limit of outer solution) [βe]
(or outer limit of inner sol^n)

$$y = \underbrace{(\alpha - \beta e)e^{-x/\xi} + \beta e^{1-x}}_{y_0} + O(\xi) + O(\varepsilon^2) + \cdots$$

Stability of dynamical systems

→ Large Bi (Distributed) → PDE

→ Small Bi (Lumped) → ODE

Non-isothermal CSTR, 1st order Reaction

(non-dimensional version)

$$\frac{d\tilde{C}}{d\tilde{t}} = -\tilde{C} + Da\,(1-\tilde{C})\,e^{\tilde{T}}$$

$$\frac{d\tilde{T}}{d\tilde{t}} = -(1+\beta)\tilde{T} + B\,Da\,(1-\tilde{C})\,e^{\tilde{T}}$$

At SS.

$$-\tilde{C} + Da\,(1-\tilde{C})\,e^{\tilde{T}} = 0$$

$$-(1+\beta)\tilde{T} + B\,Da\,(1-\tilde{C})\,e^{\tilde{T}} = 0$$

$$(1+\beta)\tilde{T} + B\tilde{C} = 0$$

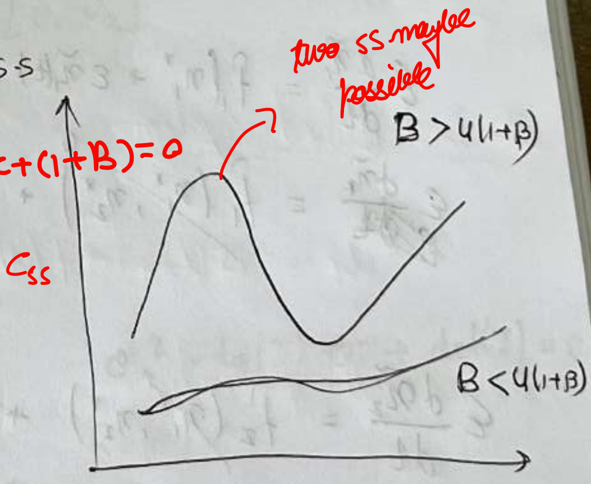
$$\tilde{T} = -\frac{B\tilde{C}}{1+\beta}$$

$$\therefore \quad Da = \underbrace{\frac{\tilde{C}}{1-\tilde{C}}\,e^{-\frac{B\tilde{C}}{1+\beta}}}_{f(\tilde{C})}$$

$$f'(\tilde{C}) = \frac{\tilde{C}}{1-\tilde{C}}\left(\frac{-B}{1+\beta}\right)e^{-\frac{B\tilde{C}}{1+\beta}} + \frac{1-\tilde{C}+\tilde{C}}{(1-\tilde{C})^2}\,e^{-\frac{B\tilde{C}}{1+\beta}} = 0$$

$$\tilde{C}(1-\tilde{C}) = \frac{1+\beta}{B} \quad SS$$

$$\tilde{C}^2 - C + \frac{1+\beta}{B} = 0 \quad \rightarrow Bc^2 - c + (1+\beta) = 0$$



$$\tilde{C}_{1,2} = \frac{+1 \pm \sqrt{1 - 4\left(\frac{1+\beta}{B}\right)}}{2}$$

$$\tilde{C}_{1,2} = \frac{B \pm \sqrt{B^2 - 4B(1+\beta)}}{2B}$$

For real roots $\quad B > 4(1+\beta)$ → To have real roots

System of ODE

$$\frac{dx_1}{dt} = \dot{x}_1 = f(x_1, x_2) \qquad \dot{x} = F(x)$$

$$\frac{dx_2}{dt} = \dot{x}_2 = f(x_1, x_2) \qquad X = [x_1 \;\; x_2]^T$$

$$F = [f_1 \;\; f_2]^T$$

I. Identify the SS.

$$f_1(x_1^*, x_2^*) = 0 \quad ; \quad f_2(x_1^*, x_2^*) = 0$$

$$F(X^*) = 0 \quad \text{when } X^* = [x_1^* \;\; x_2^*]^T$$

II. Linearisation → small perturbation added around SS of state variable (ε)

$$x_1 = x_1^* + \varepsilon \tilde{x}_1(t)$$

$$x_2 = x_2^* + \varepsilon \tilde{x}_2(t)$$

Adding perturbation

$$X^* = X^* + \varepsilon \tilde{X}(t)$$

$$\varepsilon \frac{d\tilde{x}_1}{dt} = f_1(x_1^* + \varepsilon \tilde{x}_1(t),\; x_2^* + \varepsilon \tilde{x}_2)$$

$$\varepsilon \frac{d\tilde{x}_1}{dt} = \cancel{f_1(x_1^*, x_2^*)}^{\,0} + \frac{\varepsilon x_1(t)}{1!} \left.\frac{df_1}{dx_1}\right|^* + \frac{\varepsilon \tilde{x}_2}{1!} \left.\frac{df_1}{dx_2}\right|^*$$

→ Taylor series expansion

$$\varepsilon \frac{d\tilde{x}_2}{dt} = \cancel{f_2(x_1^*, x_2^*)}^{\,0} + \varepsilon \tilde{x}_1(t) \left.\frac{df_2}{dx_1}\right|^* + \varepsilon \tilde{x}_2(t) \left.\frac{df_2}{dx_2}\right|^*$$

$$\frac{d}{dt}\begin{bmatrix} \tilde{x}_1(t) \\ \tilde{x}_2(t) \end{bmatrix} = \begin{bmatrix} \frac{df_1}{dx_1} & \frac{df_2}{dx_2} \\ \frac{df_2}{dx_1} & \frac{df_2}{dx_2} \end{bmatrix} \begin{bmatrix} \tilde{x}_1(t) \\ x_2(t) \end{bmatrix}$$

$$\boxed{\dot{\tilde{x}}(t) = \underline{\underline{J}}\,\tilde{x}(t)}$$

$$\dot{\tilde{x}}(t) = \left[\frac{d\tilde{x}_1}{dt} \quad \frac{d\tilde{x}_2}{dt}\right]^T$$

$$\underline{\underline{J}} = \begin{bmatrix} \partial f_1/\partial x_1 & \partial f_1/\partial x_2 \\ \partial f_2/\partial x_1 & \partial f_2/\partial x_2 \end{bmatrix}$$

Step III

disturbance — growth rate of disturbance

$$\left.\begin{aligned} \tilde{x}_1 &= u_1 e^{\sigma t} \\ \tilde{x}_2 &= u_2 e^{\sigma t} \end{aligned}\right]$$

most Process, if process observable time is very less than

$$\frac{dy}{dt} = (\text{const})\, y$$

nature of perturbation is 1st order

$\sigma > 0$, unstable → perturbations will grow with t

$\sigma < 0$, stable → with time disturbance will be suppressed (perturbations)

$$u_1 \sigma e^{\sigma t} = u_1 e^{\sigma t} \frac{df_1}{dx_1}\Big|^* + u_2 e^{\sigma t}\frac{df_1}{dx_2}\Big|^*$$

$$u_2 \sigma e^{\sigma t} = u_1 e^{\sigma t} \frac{df_2}{dx_1}\Big|^* + u_2 e^{\sigma t}\frac{df_2}{dx_2}\Big|^*$$

$$\sigma U = \underline{\underline{J}}\, U$$

where $U = [u_1\ u_2]^T$

$$(\sigma I - \underline{\underline{J}})\, U = 0$$

$$|\sigma I - \underline{\underline{J}}| = 0$$

$$\begin{vmatrix} \sigma - \frac{df_1}{dx_1}\Big|^* & -\frac{df_1}{dx_2}\Big|^* \\ -\frac{df_2}{dx_1}\Big|^* & \sigma - \frac{df_2}{dx_2}\Big|^* \end{vmatrix} = 0$$

$$\sigma^2 - tr(J)\,\sigma + det(J) = 0$$

$Re(\sigma) < 0$

$$\sigma = \frac{tr\, I \pm \sqrt{(tr(J))^2 - 4\,det(J)}}{2}$$

Calculate the eigen value 'σ'

all → $Re(\sigma) < 0$ → stable all → $Re(\sigma) > 0$ ⇒ unstable

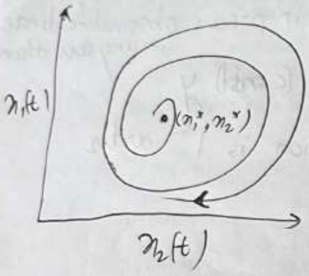
Bifurcation Analysis :-

$\sigma = s \pm i w$

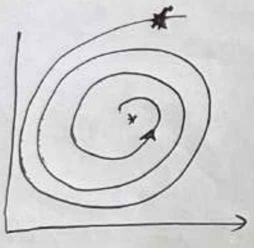
$Re(\sigma) = 0 = s$ (Hopf bifurcation) (purely imaginary) " $tr(J) = 0$

$\rightarrow$ Pitch fork

$\rightarrow$ Transcritical



$\rightarrow$ unstable $\rightarrow Re(\sigma) > 0$



stable $Re(\sigma) < 0$

Irrespective of imaginary part, real part of $\sigma < 0$ for all σ_i's $\rightarrow$ stability criterion

for $Re(\sigma) < 0$ $\quad tr(\underline{\underline{J}}) \cancel{>} 0$ & $\det \underline{\underline{J}} < 0$

Distributed system

$$u_t = D u_{xx} + f(u) \quad ; \quad u(x,t)$$

Add small perturbation, $u = u_*\big|_{x^*, t^*} + \delta u(x,t)$ $\quad$ (u_* → stable state)

$$[u_* + \delta u]_t = D[u_* + \delta u]_{xx} + f(u_* + \delta u)$$

$$\delta u_t = D\,\delta u_{xx} + \cancel{f(u_*)}^{0} + f'(u_*)\,\delta u + \underbrace{\quad\quad}_{\text{ignore}}$$

At stable state $\quad \cancel{u_{*t}}^{0} = D\,\cancel{u_{*xx}}^{0} + f(u_*)^{\rightarrow 0}$

$$\therefore \frac{\partial \delta u}{\partial t} = D \frac{\partial^2 \delta u}{\partial x^2} + \delta u\, f'(u_*)$$

$$\delta u = \alpha e^{\sigma t} e^{ikx}$$

$\rightarrow \cos(kx) + i\sin(kx)$] fourier decomposition of spatial domain

$$\alpha e^{ikx} \sigma e^{\sigma t} = -k^2 \alpha e^{ikx} e^{\sigma t} D + \alpha e^{\sigma t} e^{ikx} f'(u_*)$$

$$\sigma = -k^2 D + f'(u_*)$$

If $f = \beta u \Rightarrow f' = \beta$

$$\boxed{\sigma = \beta - Dk^2}$$

magn of rate const & wave no.

Stability

$k = 1, 2, \ldots \infty$

$\Downarrow$

for very large 'k' value

$\Downarrow$

always stable

$\frac{d\sigma}{dk} = 0 \Rightarrow$ Dominant mode $\rightarrow$ stable $\rightarrow$ Then whole system

Saffman - Taylor instability

Ryleig - Bernard Ins "

Kevin - helmoltz "

Continuum models $\rightarrow$ momentum, Energy, Mass — Conservation

$\rightarrow$ Bulk behaviour is represent

System (charachteristic) length scale $\sim$ mean free path length of the molecule.

$\rightarrow$ Continuum approximation is <u>not</u> valid

$$\frac{\lambda}{l_c} = K_n \geq 1 \longrightarrow \left\{ \begin{array}{l} \text{Rarefied gases} \\ \text{Nano scale processes} \end{array} \right] \text{Discreate Models are used}$$

Particle Level Process

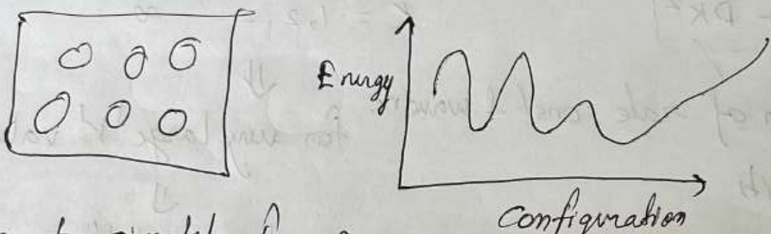
└→ Dispersed phase models → each phase is continuum domain $(Kn < 1)$

└→ Molecular dynamic models → $\boxed{Kn > 1}$

mathematical model are similar in nature by the physics are very different.

Kinetic Monte Carlo Method → Discrete dynamic Method (stochastic base method)

Monte Carlo –



We try to simulate for Random configuration

KMCM → To bring in some process dynamic into Picture of Rand configuration.

Random Numbers

↳ There is no possible correlation between the Random numbers

Random Number generator algorithms –

→ Mersenne Twister
→ SIMD-oriented fast Mersenne Twister
→ Combined multiple recursive
→ Multiplicative Lagged Fibonacci
→ philox - 4x32 generator
→ Threefry 4x64 generator

we give seed as initialization for the given algorithm
⇓
If fixed the seed we get the same random number.

Steps of KMC

1. Initialise your state $(t=0)$

2. Determine maximum possible transition states (N_K)

3. Assign probabalities to each of these transition states. such that sum is One.

4. Choose randomly a particular event (transition) to execute.

5. ~~Assign a~~ choose a Random number $\rightarrow (0,1) \rightarrow [U]$ and see if the probability of particular event (transition) is greater than Random number choosen above. If it is, then mark it down

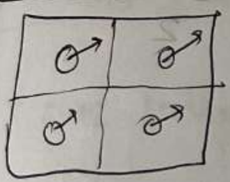
6. Change your time $t \rightarrow t + \Delta t$

$$\Delta t = (N_K \, r)^{-1} + \ln(1/U)$$

Seed should be same

r ↓ ~~No. of particle~~ possible transition rate of the event

Initialization:



[0, half cell size] 4X2 → Domain $[0, 2\pi]$

$$\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \\ a_3 & b_3 \\ a_4 & b_4 \end{bmatrix}$$

a ↓ How much we displacing the particle

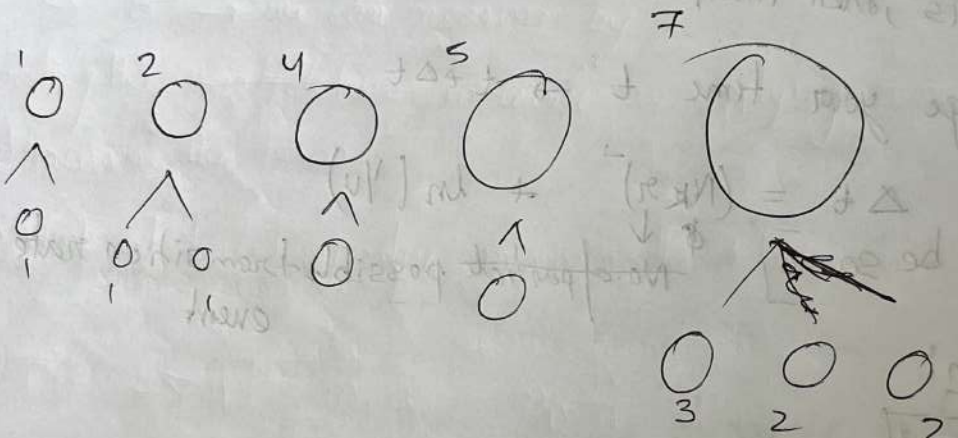
b → Direction of Movement

Metropolis - Hasting Algorithm

Tutorial Problem

Consider a system consisting of five particles of volume 1, 2, 4, 5, 7. These particles undergo breakage. The breakage rate is proportional to their volume with the proportionality constant is unity. Find the state of the system after 20 events. Use the same seed for random number generation.

Use hand calculations to estimate the particle size distribution after 2 events.



Molecular Modelling

(discrete particles) where $K_n > 1$

Molecular dynamics

↳ Interactive force

↳ Newton's law of motion

$$\vec{F} = -\nabla\phi = m_p \ddot{\vec{r}}$$

$\vec{r}$ → position; $\ddot{\vec{r}}$ → acc^n; ↳ vector

Interactive Energy

- → Bonded → Purely Chemical bonding or Physical bond exist b/w the molecule
- → Non-bonded → (Van der-walls, Columbic)

for Bonded : Bond stretching, Bond rotation in 2D, 3D, Bond Vibration — Type of bonded Interaction.

$$E_{bond\text{-}stretch} = \sum_{bonds} k_b (b - b_0)^2$$

$$E_{bond\text{-}bent} = \sum_{angl} K_\theta (\theta - \theta_0)^2 + \sum K_{urey\text{-}Bradel} (S - S_0)^2$$

S ↓ solid angle.

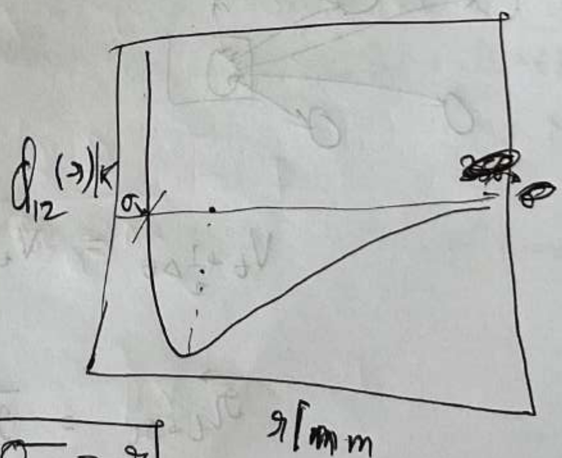
$$E_{bont\text{-}rotation} = \sum_{dihedra's} K_\chi (1 + \cos(\eta\chi - \delta)) + \sum_{impropers} k_{imp} (\phi - \phi_0)^2$$

For out of the plan.

Vander-walls interaction

LJ 6-12 interaction based on hard sphere model

$$E_{vdw} = \sum_{non\text{-}bond} 4\varepsilon \left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6} \right]$$



$\boxed{\sigma = r}$ → m

Electrostatic Interaction

↳ large scale (range)

$$E_{electrostat} = \frac{q_i q_j}{\varepsilon_d r_{ij}}$$

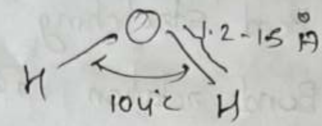
↓

$\frac{1}{r}$ → variation

$$E_{bond\text{-}stretch} = \sum_{bond} K_b (b - b_0)^2$$

measured X-ray protein crystallography

↓ measure (bond lengths, bond angle)



of the molecule

High pressure spectrum

↓ measure energy needed

(Raman Spectroscopy)

Van der -

↳ σ → we can measure by seperating the ~~particle~~ two molecule by applying / changing ~~some~~ energy / pressure and measure the forces

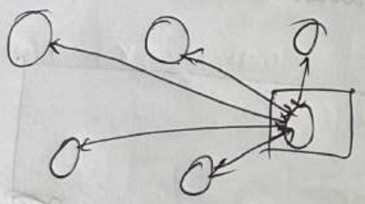
↳ work at low Temperature to minimize the Change in K.E

$$\text{K.E} = \frac{3}{2} K_B T$$

#

$$\vec{F} = -\nabla \phi = m_p a = m_p \frac{d\vec{v}}{dt} = m_p \frac{d^2 \vec{r}}{dt^2}$$

$$m_p \frac{V^{N+1} - V^N}{\Delta t} \quad \Bigg| \quad m_p \frac{r^{N+1} - 2r^N + r^{N-1}}{(\Delta t)^2}$$



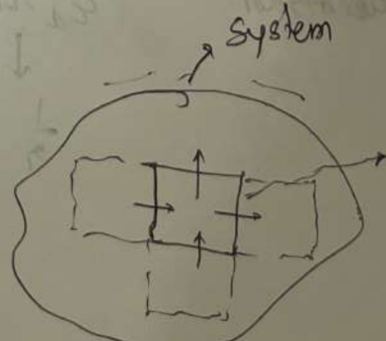
$$\left.\begin{aligned} V_{t+\frac{1}{2}\Delta t} &= V_{t-\frac{1}{2}\Delta t} + \vec{a}\,\Delta t \\ \vec{r}_{t+\Delta t} &= \vec{r}_t + \vec{V}_{t+\frac{1}{2}\Delta t}\,\Delta t \end{aligned}\right]$$ Leap-frog Verlet algo

Molecular dynamics calculation

- Periodic Boundary Condition

Ex- System → 1m × 1m × 1m

↳ System for consideration for simulation purpose



1 μm × 1 μm × 1 μm (small enough

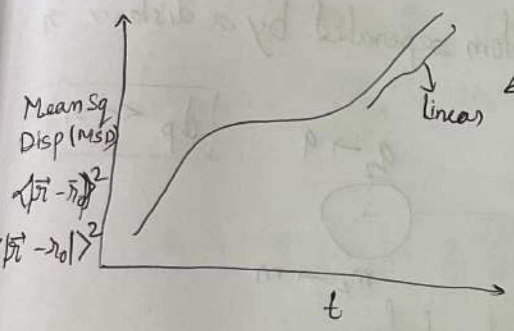
Ergodic Hypothesis : Temporal average for a small No. of mp. system for a very long time is same as spatial / ensemble average of the large no. of particles for a short period of time.

$$\langle x \rangle = \lim_{\tau \to \infty} \frac{\int_0^{\tau} x(t)\,dt}{\tau} = \lim_{M \to \infty} \frac{1}{M} \sum_{i=1}^{M} x_i$$

* No matter where your system is started, it can get to another point in phase space

Diffusivity

→ Displacement profile is known



← We have to apply Isothermal fluctuations

↓

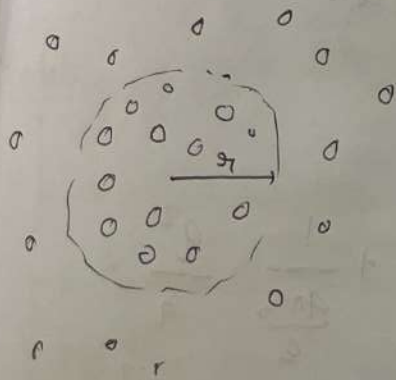
$\langle K.E \rangle = \frac{3}{2} k_B T$

$$\lim_{t \to \infty} MSD = 6Dt$$

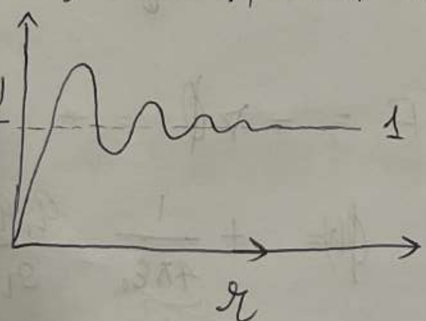
$\boxed{3D}$ case → consider Hard sphere model → $D = \frac{6\pi\mu r_p}{\vartheta}$

Radial distribution function (RDF)

By counting the no. of particle / Volume of domain r



$$g(r) = \frac{\rho(r)}{\rho_0}$$



$$\lim_{r \to \infty} g(r) \to 1$$

Average density at 'r' → $\rho(r) = \rho_0\, g(r)$

Compressibility $\rightarrow$ $K_T = \frac{1}{\rho k_B T}\left[1 + \rho \int_0^\infty (g(r) - 1) dr\right.$

Excess internal energy $\rightarrow$ $E^{ex} = \frac{\rho^2 V}{2} \int_0^\infty V(r) g(r) dr$

Helmontz free energy $\rightarrow$ $w(r) =$

memory intensive, we ~~used~~ large no. particle for short spatial average

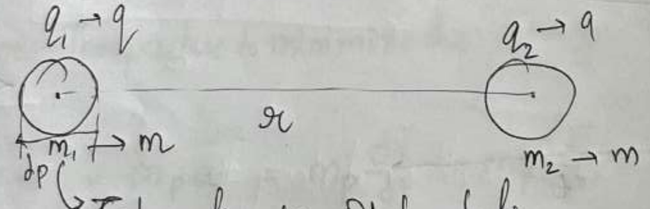
KMC $\rightarrow$ stochastic based approached, initial states does not matter.

Two differences $\rightarrow$

MD $\rightarrow$ Deterministic model, initial states matter, time intensive calculation, use temporal averag

Problem

(13) (b) Consider a system of two charged atom seperated by a distance 'r'



$\boxed{d_p << r}$

Interaction is Electrostatic.

Derive expression trajectory & velocity wrt time.

$$F = \frac{q_1 q_2}{4\pi \varepsilon_0 r^2} = \frac{q^2}{4\pi \varepsilon_0 r^2}$$

$$F = -\nabla \phi =$$

$$\phi(r) = +\frac{1}{\underbrace{4\pi\varepsilon_0}_{\varepsilon}} \frac{q_1 q_2}{r} = +\frac{1}{\underbrace{4\pi\varepsilon_0}_{\varepsilon}} \frac{q^2}{r}$$

$$\vec{F} = m_p \vec{a} \Rightarrow \frac{q_1 q_2}{\underbrace{4\pi\varepsilon_0}_{c} r^2 m_p} = \vec{a} = \frac{q^2}{\underbrace{4\pi\varepsilon_0}_{\varepsilon_0} r^2 m_p}$$

$$\cancel{V_{t+\frac{\Delta t}{2}} = V_{t-\frac{\Delta t}{2}} + \vec{a}\,\Delta t}$$

$$\cancel{\frac{d\vec{V}}{dt} = \vec{a}}$$

$$\cancel{\vec{V} = \frac{q^2}{4\pi\epsilon_0}}$$

$$m_p \frac{d^2 r}{dt^2} = + \frac{q^2}{m_p \epsilon r^2}$$

$$\frac{d^2 r}{dt^2} = \frac{\alpha}{r^2}$$

$$r = CF + PI$$

$$CF : \ddot{r} = 0 \Rightarrow r = C_1 t + C_2$$

$$PI : r_{PI} = \left(\frac{9}{2}\alpha\right)^{1/3} t^{2/3}$$

$$\therefore \quad r = C_1 t + C_2 + \left(\frac{9}{2}\alpha\right)^{1/3} t^{2/3}$$

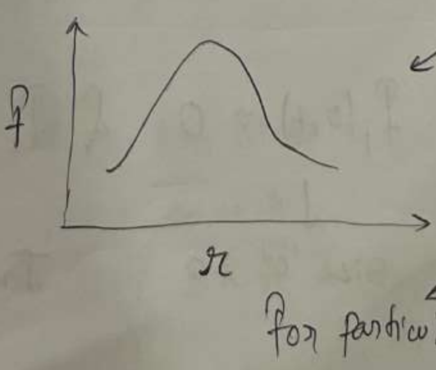
$$\text{Velocity} : \dot{r} = C_1 + \left(\frac{9}{2}\alpha\right)^{1/3} \frac{2}{3t^{1/3}}$$

$$\frac{d r^2}{dt^2} = -\frac{\alpha}{r^2} \longrightarrow \text{Emdem fowler equation.}$$

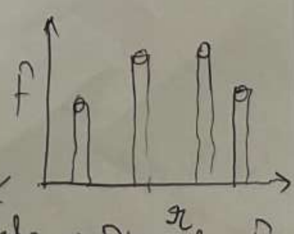
Modelling of particulate / dispersed phase

- → Crystallisation
- → Aerosol formation
- → Bubbly flows
- → Liq - Liq Emulsion

Countable # particle / entity



For continuum → a continuous fun.



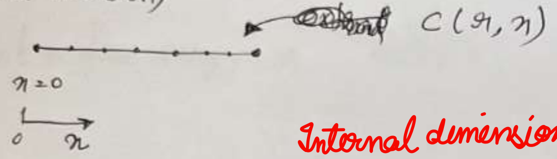
For particulate → Discrete fn

Co-ordinates (Independent vars)

→ external : spatial position, time — properties of system

→ Internal : <u>size</u> , <u>mass</u> etc. → related "to" the particle "Entity characteristics"

(extra-dimension)

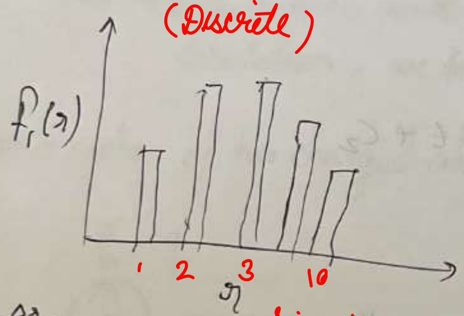


Internal dimension is treated same way as external

Due to internal variable, it adds extra variable/dimension to the external co-ord.

$$f_1(r) = \frac{\#\text{ particle}}{\text{Total }\#\text{ particle}} = \text{Number density}$$

(Discrete)



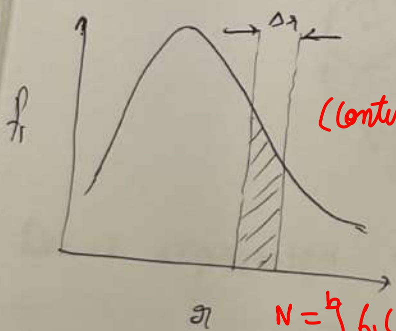
$$f_1(r,t)\,\Delta r\big|_t - f_1(r,t)\,\Delta r\big|_{t+\Delta t}$$

$$= \Big[f_1(r,t)\,G(r,t)\big|_r - f_1(r,t)\,G(r,t)\big|_{r+\Delta r} \Big]\Delta t$$

$$\frac{\partial f_1(r,t)}{\partial t} + \frac{\partial}{\partial r}\Big[f_1(r,t)\,G(r,t)\Big] = 0$$

↓

Population balance equation



(continuous)

\# particles in this Δr.

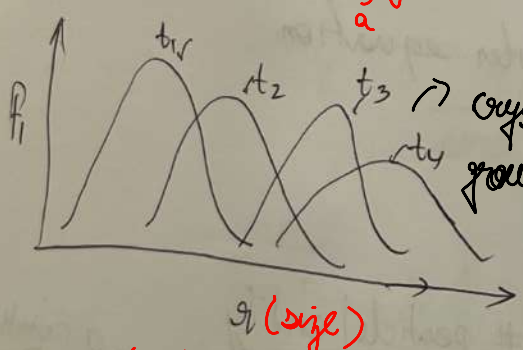
$$f_1(r,t)\,\Delta r\big|_{t+\Delta t} - f_1(r,t)\,\Delta r\big|_t$$

$$N = \int_a^b f_1(r,t)\,dr = \Big[\underbrace{\#\text{ particle}\big|_r} - \underbrace{\#\text{ particle}\big|_{r+\Delta r}}\Big]$$

$$\Delta t\Big[f_1(r,t)\,G(r)\big|_r - f_1(r,t)\,G(r)\big|_{r+\Delta r}\Big]$$

↑

Growth rate



$t_1 < t_2 < t_3 < t_4$

$$\underbrace{\frac{\partial f_1(r,t)}{\partial t} + \frac{\partial}{\partial r}\Big[G(r)\,f_1(r,t)\Big]}_{\text{Population balance eq}^n} = \underset{\text{(source)}}{h^+} - \underset{\text{(sink)}}{h^-} = 0$$

No source sink s'

$$f_1(0,t) = 0 \quad \& \quad f_1(r,0) = N_0$$

↓ size '0' = 0

⇓ Initial timed ⇒ Seeded

Typically for seeded crystallization (homogenous)

$G f_1 = \dot{N}_0$ (initial nucleation)

$$\dot{N}_0 = K_1 \underbrace{\Phi(c)}_{\text{collision freq}} \exp\left[-\frac{K_2}{\{\ln(c/c_s)\}^2}\right]$$

c_s → Saturation conc

Particle breakage

↳ droplet breakup

↳ polymer size distri

↳ Binary fission

h^+ & h^-] → Both will be present

h^+ ↓ net birth rate

h^- ↳ net death rate

$\underbrace{h^+ - h^-}_{\text{Net Birth rate}}$

Breakage functions

1. Breakage Frequency : $K(r,t)$

2. Avg. # new particles formed ; $\nu(r',t)$ integer fn. → After breakage event

minimum 2, integer always (parent) x' → x (daughter) $[x < x']$

3. daughter size distribution, $P(r|r',t)$

$$\int_0^\infty P(r|r',t)\,dr = 1 \rightarrow \int_0^{r^*} P(r|r',t)\,dr = 1$$

if $m(r') \geq m(r^*) \rightarrow P(r|r',t) = 0$ No daughter size beyond r' possible

r' → new particle size

r → parent size

$$m(r^*) \geq \nu(r^*,t) \int_0^{r^*} m(r')\, P(r|r',t)\, dr'$$

~~$h^- = K f_1(r',t)$~~ $\quad h^- = K f_1(r,t)$ → sink

$r < r' < \infty$

$$h^+ = \int_{r'}^{\infty \rightarrow r} \nu K f_1 P \,dr' = \int_{r'}^{r} \nu K f_1 P\, dr'$$

$\because P(r|r',t) = 0$ if $r' > r$

Crystal growth

$\frac{dr}{dt} = G(r)$

Diffusion control on growth

$$\frac{dc}{dr} = \frac{c^* - c_b}{r}$$

c^* → saturation conc.

c_b → bulk conc.

$V_s = \frac{4}{3}\pi r^3$

$$\frac{d}{dt}(\rho V_s) = D(4\pi r^2)\frac{\Delta c}{r}$$

$$\rho \frac{d}{dt}\left(\frac{4}{3}\pi r^3\right) = D 4\pi r^2 \frac{\Delta c}{r}$$

$$\frac{dr}{dt} = \frac{K}{r}$$

$\frac{dr}{dt} \sim \frac{1}{r}$ → diffusion controlled process

$\frac{dr}{dt} \sim$ constant → for surface controlled

(left margin, partially visible from previous page: ...ternal co-ord, $^+$ h^-, $-h$, ...) (sink), ...ound, sink, s')

Dispersed Phase - Model

- → Number density function
- → PBE
- → Breakage (source & sink)
- → Aggregation (aerosol dynamics)
  - ↳ Ex :- cloud Formation
    mist , droplets
    Sintering

Aggregation → Joining of two particles

↳ Aggregation frequency
&
Pair Correlation (distribution) function

$f_1(x)$ → Number density function

$f_2(x, x')$ → distinct pair density function which are aggregating.
(two particles)

↳ Binary (✓)

f_3, f_4] Triplet , Quadruplet (✗)

$\underbrace{(x, r) \; \& \; (x', r')}_{\text{pair of particles}} \rightarrow \underbrace{(\tilde{x}, \tilde{r})}_{\text{New particle that is formed}}$

particles which are aggregating

$x , x' \longrightarrow f_1(x) , f_1(x')$] Number density function

new state, $\tilde{x} \longrightarrow f_1(\tilde{x})$

$(x, x' < \tilde{x})$

$a(x, r ; x', r' ; t)$ → aggregating frequency

PBE : RHS → h^{+} + h^{-1}

(new state) (source) ; Sink (old state)

$a(x, r ; x', r' ; t)$
$= a(x', r' ; x, r ; t)$
⇓
Symmetry for binary agg.

$$h^{+} = \int_{\Omega x} dV_x \int_{\Omega r} \frac{1}{\delta}\, dV_r\; a(\tilde{x},\tilde{r};x',r';t)\, f_2(\tilde{x},\tilde{r}:x',r',t) \boxed{\frac{\partial(\tilde{x},\tilde{r})}{\partial(x,r)}}$$

$h^{+}(x,r,t)$

Duplicating factor → represents no. of times identical pairs have been considered

relative factor relating new state which will be the kind of old state for next

$$a(\tilde{x},\tilde{r};x',r';t) = a(\tilde{x},\tilde{r};x,r;t)$$

$$\frac{\partial(\tilde{x},\tilde{r})}{\partial(x,r)} = \begin{bmatrix} \frac{\partial\tilde{x}_1}{\partial x_1} & \frac{\partial\tilde{x}_2}{\partial x_2} & \cdots & \frac{\partial\tilde{x}}{\partial x_n} & \frac{\partial\tilde{x}_1}{\partial r_1} & \frac{\partial\tilde{x}_1}{\partial r_2} & \frac{\partial\tilde{x}}{\partial r_3} \\ \vdots \\ \frac{\partial\tilde{x}_n}{\partial x_1} \\ \frac{\partial\tilde{r}_1}{\partial x_1} \\ \vdots \\ \frac{\partial\tilde{r}_2}{\partial x_1} \end{bmatrix}$$

No $\tilde{x},\tilde{r}$

$$h^{-1}(x,r,t) = \int_{\Omega x} dV_x \int_{\Omega r} dV_r\; a(x',r';x,r;t)\, f_2(x',r';x,r,t)$$

~~$h^{+}(\tilde{x},\tilde{r},t) = \int dV_x \int dV_r\; a(x,r;\tilde{x},\tilde{r},t)\; f_2(x,r;\tilde{x},\tilde{r};t)$~~

Pair function : $\cancel{f_2(x,r;\tilde{x},\tilde{r};t)}$ $f_2(x',r';x,r;t) = \underbrace{f_1(x,r,t)\; f_1(x',r',t)}_{\text{approximation}}$

Aerosol dynamics → dispersion of solid in fluids

Range of particle size, 'nm → mm'
6 ords of magⁿ variation

Divided into two parts
- → Discrete (small size Range) $\boxed{x_i = i\,x_1}$ → $x_0 < x < x_n$
  ↳ $(1 < i < n)$ ↳ integer multiples. like particles with size 5 nm, 10 nm, 15 nm and multiple of smallest
- → Continuous → $x_n < x < \infty$

↱ i in both density and size for discrete

$$N(t) = \sum_{i=1}^{n} f_{1,i}(t) + \int_{x_n}^{\infty} f_1(x,t)\, dx$$

Note

1. There could be jump from discrete range to continuous range particle
2. Aggregating frequency : a_{ij} (discrete ranges : x_i & x_j)

$a(x, x')$ [continuous range]

$a_i(x)$ [one particle from discrete $\rightarrow x_i$, one particle in continuous $\rightarrow x$]

③ There is no growth. ④ Lumped system

⑤ Homogeneous generation [removal of particles both in discrete & continuous range]

PBE : $\frac{\partial f_{1,i}(t)}{\partial t} = h_i^+ - h_i^-(t)$ [Discrete Range]

$\frac{\partial f_1(x,t)}{\partial t} = h^+(x,t) - h^-(x,t)$ [continuous Range]

Discrete Range:

$i=1$

$$h_1^+(t) = \sum_{j=2}^{n} \overset{\text{evaporation rate}}{e(x_j)} f_{1,j} (1 + \overset{\text{Kronekar delta}}{\delta_{j,2}}) + \int_{x_n}^{\infty} e(x) f_1(x,t)\, dx$$

→ Smallest particle can only happen due to evaporation, not because of aggregation

accounts for double contribution by particle x_2

$$\delta_{n,m} = 1 \; ; n = m$$
$$= 0 \; ; m \neq m$$

$2 < i < n \rightarrow$

$$h_i^+(t) = \frac{1}{2} \sum_{j=1}^{i-1} a_{i-j,j} \underbrace{f_{1,i-j} f_{1,j}}_{f_2(i-j,j)} + \underbrace{\delta_{in} \int_{x_n}^{x_n + x_1} e(x) f_1(x,t)\, dx}_{\text{evap. from continuous}}$$

$$+ \underbrace{e(x_{i+1}) f_{1,i+1}}_{\text{eva from discrete}}$$

↓ considering binary

aggreg.

condn → evaporation

$$h_i^-(t) = f_{1,i} \left\{ \sum_{j=1}^{n} a_{i,j} f_{1,j} + \int_{x_n}^{\infty} a_i(x) f_1(x,t)\, dx \right\}$$

$$+ e(x_i) f_{1,i}$$

$$\delta_{i,n} = \begin{cases} 1 & \text{if } i = n \\ 0 & i \neq n \end{cases}$$

A particle can be generated by
a) evaporation of large particle
b) Aggregation
c) External generation

Nothing else

$a(x, x')$ [continuous range]

$a_i(x)$ [one particle from discrete, one particle in continuous]

(3) There is no growth. (4) Lumped system

S

PBE : $\dfrac{\partial f_{1,i}(t)}{\partial t} = h_i^+ - h_i^-(t)$ [Discrete Range]

$\dfrac{\partial f_1(x,t)}{\partial t} = h^+(x,t) - h^-(x,t)$ [Continuous Range]

Discrete Range:

$i=1$

$$h_1^+(t) = \sum_{j=2}^{n} e(x_j) f_{1,j} (1 + \overset{\text{Kronecker delta}}{\delta_{j,2}}) + \int_{x_n} e(x) f_1(x,t) dx$$

→ Smallest particle can only happen due to evaporation, not because of aggregation

$1<i<n$ → δ

$$\delta_{n,m} = 1 \quad ; n = m$$
$$= 0 \quad ; m \neq m$$

$$2<i<n \rightarrow h_i^+(t) = \frac{1}{2}\sum_{j=1}^{i-1} \overbrace{a_{i-j,j} \underbrace{f_{1,i-j} f_{1,j}}_{f_2(i-j,j)}}^{\text{aggrega}} + \overbrace{\delta_n \int_{x_n}^{x_n+x_1} e(x) f_1(x,t) dx}^{\text{condn} \rightarrow \text{evaporation}}$$

↓ considering binary

$$+ \underbrace{e(x_{i+1}) f_{1,i+1}}_{\text{eva from discrete}}$$

$$h_i^-(t) = f_{1,i}\left\{ \sum_{j=1}^{n} a_{i,j} f_{1,j} + \int_{x_n}^{\infty} a_i(x) f_1(x,t) dx \right\}$$

$$+ e(x_i) f_{1,i}$$

Continuous range: $x_n \leq x < \infty$

$$\frac{\partial f_1(x,t)}{\partial t} = h^+(x,t) - h^-(x,t)$$

For Discrete to joint go continuous range —

$x_i + x_j = x_{i+j}$ where $i+j > n$

↓ what should be the mass

$x_{i+j} < x < x_{i+j} + x_0$ → size range

we distribute the mass which formed $(1/x_0)$ ↳ to the distribution in continuous.

$$h^+(x,t) = \underbrace{\left[\frac{1}{2x_0}\sum_{j=\left[\frac{x}{x_0}\right]-n}^{n} a_{[x/x_0]-j,\,j}\, f_{1,[x/x_0]-j}\, f_{1,j}\right]\left(H(x - x_{[x/x_0]}) - H(x - x_{[x/x_0]+1})\right) \times \left(1 - H(x - x_{2n})\right)}_{\text{Discrete}}$$

(source)

$$+\left[H(x - x_{n+1})\sum_{j=1}^{n} a_j(x - x_j)\, f_{1,j}\, f_1(x - x_j, t)\right] \rightarrow \text{contin + discr}$$

$$\boxed{H(x - x_{i+j}) - H(x - x_{i+j+1}) = \begin{cases} 1 \; ; & x_{i+j} < x < x_{i+j} + x_0 \\ 0 \end{cases}}$$

↳ Heavy side function

$$+\frac{1}{2}\int_{x_n}^{\infty} a(x', x - x')\, f_1(x', t)\, f_1(x - x', t)\, dx \Big] \text{ cont + cont}$$

$$+\frac{1}{2}\, e(x + x_0)\, f_1(x + x_0, t)\Big] \text{ evaporation.}$$

$$h^-(x,t) = f_1(x,t)\left[\sum_{j=1}^{n} a_j(x)\, f_{1,j} + \int_{x_n}^{\infty} a(x, x')\, f_1(x', t)\, dx'\right]$$

(Sink)

$$+\, e(x)\, f_1(x,t)$$

$f_1(0) = u_i$ when $i = 0, 1, 2 \ldots \infty$

$f_1(x, 0) = u(x) \quad \forall\, x \geq x_n$

Typical Aggregation Problem

$$\frac{\partial f_1(x,t)}{\partial t} = -f_1(x,t)\int_0^\infty f_1(x',t)\, a_0\, dx' + \frac{1}{2}\int_0^{x^*} a_0 \underbrace{f_2(x',t)}_{f_1(x',t)\, f_1(x-x',t)} dx'$$

Initial condition, $f_1(x,0) = N_0$

$\boxed{\tau = a_0 N_0 t}$ → Non-dimensional

$$\frac{\partial f_1(x,t)}{\partial \tau} = -\frac{f_1(x,t)}{N_0}\int_0^\infty f_1(x',t)\, dx' + \frac{1}{2}\int_0^{x^*} \frac{f_1(x-x',t)}{N_0} f_1(x',t)\, dx'$$

$$f = f_1/N_0$$

$$\frac{\partial f(x,t)}{\partial \tau} = -f(x,t)\int_0^\infty f(x',t)\, dx' + \frac{1}{2}\int_0^{x^*} f(x-x',t)\, f(x',t)\, dx'$$

$$\boxed{\eta(\tau) = \frac{N(t)}{N_0} = \frac{1}{N_0}\int_0^\infty f_1(x,t)\, dx = \int_0^\infty f(x,t)\, dx}$$

If we integrate both side —

LHS

$$\int_0^\infty \frac{\partial f(x,t)}{\partial \tau}\, dx = \frac{d}{d\tau}\underbrace{\int_0^\infty f(x,t)\, dx}_{\eta} = \frac{d\eta}{d\tau}$$

RHS

$$-\int_0^\infty f(x,t)\,\eta(\tau)\, dx = \eta^2 \qquad \int_0^{x'} + \int_{x'}^\infty \rightarrow \text{In this case } \underbrace{f(u,t)}_{\text{const}}$$

(0)

$$\therefore \frac{d\eta}{d\tau} = -\eta^2 + \frac{1}{2}\int_0^\infty dx \int_0^{x^*} f(\underbrace{x-x'}_{u},t)\, f(x',t)\, dx'$$

Let $u = x - x'$ | $x \to x' \rightarrow u = 0$, $x \to \infty \rightarrow u \to \infty$

$$\frac{d\eta}{d\tau} = -\eta^2 + \frac{1}{2}\underbrace{\int_0^\infty du\, f(u,t)}_{\eta}\underbrace{\int_0^\infty dn' f(n',t)}_{\eta}$$

$$\frac{d\eta}{d\tau} = -\eta^2 + \frac{1}{2}\eta^2 = -\frac{1}{2}\eta^2$$

$$\int_{u=0}^{\infty} du = \int_{n'}^{\infty} dn'$$

$$-\frac{d\eta}{\eta^2} = \frac{d\tau}{2}$$

$$\frac{1}{\eta} - 1 = \frac{1}{2}\tau$$

$$\boxed{\eta(0) = 1}$$

$$\frac{1}{\eta} = \frac{\tau+2}{2}$$

$$\boxed{\eta = \frac{2}{\tau+2}}$$

$$\eta = \frac{1}{N_0}\int_0^\infty f_1(n,t)\, dn'$$

Laplace Transform

$$\frac{\partial f}{\partial t} = -f(n,t)\int_0^\infty f(n',t)\,dn' + \frac{1}{2}\int_0^{n} f(x-n',t)\, f(n',t)\, dn'$$

Laplace

$$\mathcal{L}(f(n,t)) = \int_0^\infty e^{-sn} f(n,t)\, dn \equiv \bar{f}(s,t)$$

↳ n-domain

$$\frac{\partial \bar{f}(s,t)}{\partial\tau} = -\eta(\tau)\,\bar{f}(s,t) + \frac{1}{2}\int_0^\infty e^{-sn}\, dn \int_0^{n} f(n-n',t)\, f(n',t)\, dn'$$

$$+\frac{1}{2}\int_{x'=0}^{\infty} dx' \int_0^{\infty} du\, e^{-s(x'+u)} f(u,t)\, f(x'$$

$$\| $$

$$\frac{1}{2}\left(\bar{f}(s,t)\right)^2$$

$x - x' = u$

$x = u + x'$

$$\boxed{\frac{\partial \bar f}{\partial \tau} = -\eta \bar f + \frac{1}{2}\bar f^2} \rightarrow \frac{2}{2+\tau}$$

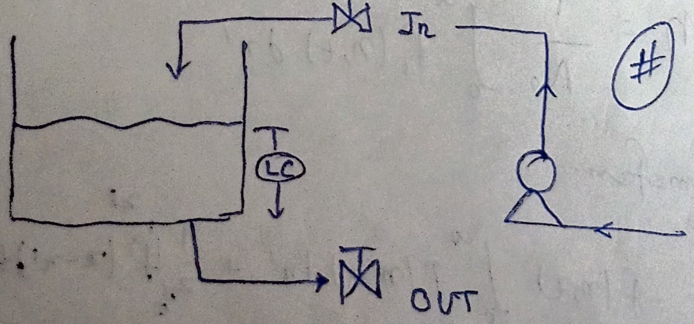
$$\bar f = \frac{1}{(2+\tau)^2}\cdot\frac{1}{C_1 + \frac{1}{2(\tau+2)}}$$

in' ↙

$f_1(x,t)$ $\quad C_1 \equiv$ eval from $\bar f(s,0)$

$$N_0 = \int_0^\infty f_1(x,0)\,dx \quad ; \quad f_1(x,0) = g(x) \rightarrow \bar g(s)$$

Programmable Logic Controller (PLC)



States or Modes of PLC

→ Steady

→ Temporal

→ Hybrid

Logic state

1 → ON

0 → OFF

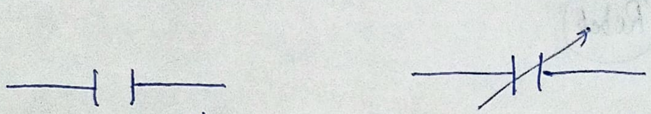
→ (4 - 20 mV) mA

1 → ON / TRUE / contact closure / open / energise etc

0 → OFF / False / contact open / closed / de-energise etc.

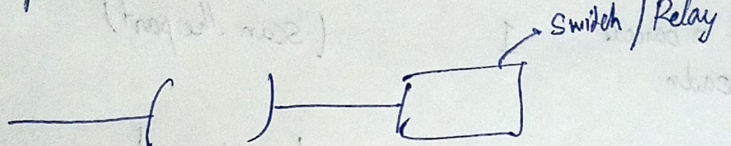
Programming Representation

Contact



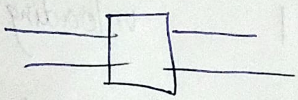
open (off) closed (ON)

Coil (output states)



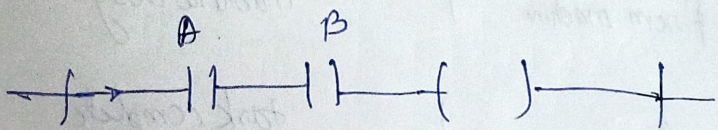
energised when power / current flows.

Boxes

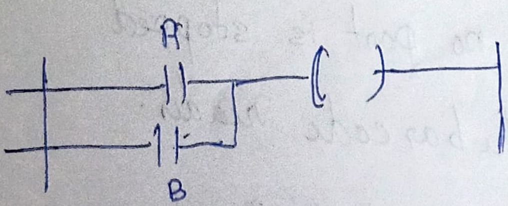


Contain function which are executed when Power / current flow through them.

AND

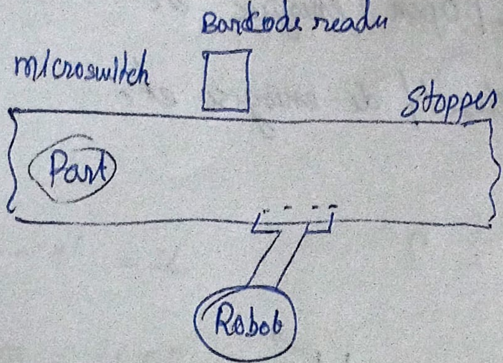


OR



↓

A1



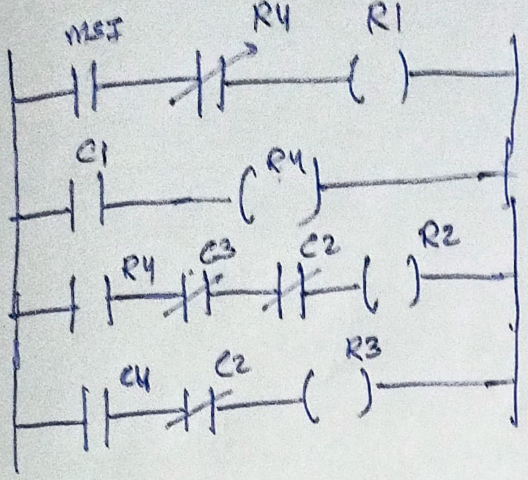
id	desc	State	
MSI	microswitch	1	(part arrive)
R1	output to barcode reader	1	(scan the part)
C1	input from barcode reader	1	(right part)
R2	output robot	1	loading cycle
R3	output ″	1	unloading cycle
C2	input from robot	1	robot busy
R4	output to stopper	1	stopper up
C3	input from machine	1	machine busy
C4	″	1	task complete

Rung 1: part arrives & no part is stopped trigger the barcode reader.

Rung 2: If part arrives, actuate the stopper

Rung 3: stopper is up, machine is not busy, Robot is not busy, load onto the machine

Rung 4: Task complete, robot is not busy, unloaded the machine.



Ladder Diagram.